What is the influence of knee joint movement on the maximal average mechanical power output of human quadriceps femoris muscle?

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Access the code and full analysis notebook at https://edwinreuvers.github.io/publications/hq-ampo

Abstract

Performance during motor activities such as wheelchair riding, cycling, rowing and speed skating, depends critically on the average mechanical power output (AMPO) produced by the muscles. To maximise short-duration performance, limb movements should allow muscles to deliver maximal AMPO. However, it is unclear which movement maximises AMPO of human muscle. In this study, we employed a Hill-type muscle-tendon-complex (MTC) model to predict the maximally attainable AMPO of human *m. quadriceps femoris* for various imposed periodic knee joint movements. Based on these predictions, we selected one set of conditions predicted to yield identical maximally attainable AMPO despite substantial variations in knee joint movements and another set of conditions predicted to yield substantial variations in maximally attainable AMPO. In the experiment, periodic knee joint movements were fully imposed by a knee dynamometer. Participants were instructed to maximise AMPO and, to this end, received visual feedback on their cumulative mechanical work throughout each cycle. Experimental data closely matched predictions derived from the Hill-type MTC model, confirming the validity of the model. Model predictions showed a substantial influence of knee joint movement on the maximally attainable AMPO. Specifically, predictions revealed a strong interaction between cycle frequency and knee joint excursion: increasing one necessitates a decrease in the other to maximise AMPO. Even more interestingly, *m. quadriceps femoris* should spend about 80% of the cycle duration while shortening, independent of cycle frequency and/or knee joint excursion.

1 Introduction

Basic motor activities such as walking, running, swimming, and activities involving man-made appliances like wheelchair riding, cycling, rowing, and speed skating are characterised by repetitive limb movements that result in periodic stretch-shortening cycles (SSCs) of muscles. In each SSC, muscles alternate between concentric and eccentric phases: they deliver positive mechanical power when generating force while shortening, and they deliver negative mechanical power when generating force while lengthening. In many motor activities, performance largely depends on the average mechanical power output (AMPO) over a full periodic cycle of all muscles involved. Maximising performance therefore requires the muscles to maximise average positive mechanical power during shortening and to minimise average negative mechanical power during lengthening. This raises the question: what periodic muscle length/velocity over time should be imposed in order to maximise AMPO?

A muscle consists of contractile fibres (the contractile element; CE) and connective tissue, both parallel to the fibres (the parallel elastic element; PEE) and in series with the fibres (the series elastic element; SEE). These three elements together form the muscle-tendon complex (MTC). Because connective tissue behaves predominantly elastically, AMPO of the MTC is determined almost exclusively by AMPO of the CE. However, CE length depends on MTC length and deformation of the elastic elements, which in turn depends on MTC force. Consequently, and ultimately, MTC AMPO depends not only on stimulation over time but also on MTC length over time.

During limb movement, MTC length varies with the joint(s) it spans. Understanding how periodic MTC length changes affect AMPO thus provides insight into how joints should move to maximise AMPO. These insights are not only interesting from a fundamental scientific perspective, but also from an applied sciences perspective, as they can inform the (re-)design of equipment. Such a design approach, that capitalises on a thorough understanding of the human musculoskeletal system has proven effective, as demonstrated by the Klapskate for speed skating (Ingen Schenau et al., 1996) and the SuperSeat for rowing (Soest and Hofmijster, 2009). Yet, such design approaches remain rare due to limited understanding of how changes in MTC length over time affect AMPO, or other measures of performance in human muscle.

Much of the existing knowledge of the influence of CE/MTC length over time on AMPO is inferred from experiments on isolated animal muscle. In these experiments, CE or MTC length and muscle stimulation are imposed and resulting AMPO is measured. This method is often referred to as the 'workloop technique' (Josephson, 1985). These experiments have often been performed for sinusoidal SSCs, showing that the effects of cycle frequency and CE/MTC length excursion interact: at lower cycle frequencies, length excursions should be larger in order to maximise AMPO. However, sinusoidal SSCs by definition result in equal shortening and lengthening durations, leaving unanswered how unequal shortening and lengthening duration affects AMPO.

Askew & Marsh (1997; 1998) were the first, and until very recently (see Reuvers et al., 2025), the only researchers to perform experiments on the influence of unequal shortening and lengthening duration

on the maximally attainable AMPO of muscle during short-duration, all-out (sprint) conditions. In their study, Askew & Marsh (1997; 1998) imposed SSCs with near-constant shortening and lengthening velocities and parametrised them in terms of three variables: cycle frequency, FTS (fraction of the cycle time spent shortening) and length excursion (see Figure 1). Askew & Marsh (1997; 1998) examined three distinct FTS values: 0.25, 0.50 and 0.75. They found that AMPO was consistently highest for FTS = 0.75 at all tested cycle frequencies at the optimal length excursion. In a recent study (Reuvers et al., 2025), we further examined the effect of SSC parameters on the maximally attainable AMPO by combining experiments on rat *m. gastrocnemius medialis* with predictions using a Hill-type MTC model. Our results showed a near-perfect correlation between model-based predictions and experimental results ($r^2 > 0.98$). The effect of cycle frequency, FTS and length excursion on the maximally attainable AMPO was then further examined using the Hill-type MTC model for a broad range of SSC parameters. These model-based predictions showed that the optimal FTS ranged 0.84-0.88 across a wide range of cycle frequencies and length excursions.

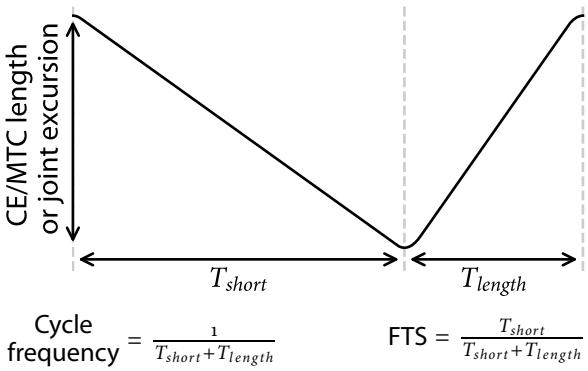

Figure 1: Parameterisation of stretch-shortening cycles. T_{short} and T_{length} denote the shortening and lengthening durations. FTS denotes the fraction of the cycle time spent shortening. In the example shown, the shortening duration is 65% of the cycle duration (i.e. FTS = 0.65).

Although animal studies have provided valuable insights, it remains unknown how these findings translate to human muscle. For example, is the optimal FTS for human muscle also in the order of 0.85? Although similar experiments can, in principle, be designed for human muscle, inherent differences exist between experiments involving human participants and experiments on isolated animal muscles. First, MTC properties can be accurately estimated in animal experiments based on dedicated experiments (see Reuvers and Kistemaker, 2025) that are challenging in human experiments (i.e. 'quick-release' experiments). Second, CE/MTC length is under full control in animal experiments, whereas in human movement it can only be imposed at the joint level. As a result, the exact CE/MTC length over time is uncertain in human experiments due to inter-individual variation in, for example, tendon compliance and moment arms. Third, stimulation is also under full control of the experimenter in animal experiments, but not in human experiments. Despite these differences, it is feasible to design human experiments in which imposed joint movements are systematically varied, with participants aiming to maximise AMPO. Similarly to animal experiments where CE/MTC length is under full control, an experimental

design should be used in which joint angle over time is fully imposed, such that the behaviour of the participants does not have any influence on the kinematics (i.e. knee joint angle over time). Such an approach allows investigation of the effects of SSC parameters – cycle frequency, FTS, and joint excursion – on maximally attainable AMPO. Surprisingly, to our knowledge such studies have not yet been conducted for human muscle – not even for sinusoidal SSCs. Consequently, the effect of SSC parameters on the maximally attainable AMPO in human muscle is currently unknown.

In this study, we examine the influence of SSC parameters on the maximally attainable AMPO of human *m. quadriceps femoris* during short-duration, all-out (sprint) conditions. Under such conditions, performance is determined by the mechanical properties of the MTC. Hill-type MTC models have been shown to accurately predict mechanical behaviour under these conditions (e.g. Ingen Schenau et al., 1988; Lemaire et al., 2016; Reuvers et al., 2025; Sandercock and Heckman, 1997; Wakeling and Johnston, 1999; Zandwijk et al., 1996). We therefore combined model predictions with *in vivo* knee dynamometer experiments. In both the simulations and the experiments, knee joint angle over time was fully imposed; in other words, the knee joint kinetics did not influence the knee joint movement. This allowed us to examine the direct influence of SSC parameters on the maximally attainable AMPO. The model was used to predict the maximally attainable AMPO across a wide range of imposed knee joint movements, from which nine insightful experimental conditions were selected. In the experiment, knee joint movements were fully imposed by a knee dynamometer, while participants aimed to maximise AMPO. Measured AMPO matched the model predictions reasonably well, confirming that the Hill-type MTC model reliably captures the influence of SSC parameters on the maximally attainable AMPO. This justified using the model to systematically examine the overall influence of imposed knee joint movement on the maximally attainable AMPO of human *m. quadriceps femoris*.

2 Methods

In this study, we examined the influence of imposed SSC parameters – cycle frequency, FTS, and joint excursion – on the maximally attainable AMPO of human *m. quadriceps femoris*. Experiments were combined with simulations and optimisations using a Hill-type MTC model. Based on model predictions, we selected one set of conditions expected to yield similar AMPO despite substantial variations in knee joint movements, and another set expected to yield large differences in AMPO. In the experiment, knee joint movements were fully imposed by a dynamometer. Participants were instructed to maximise AMPO and, for this purpose, received visual feedback on cumulative mechanical work throughout each cycle. Measured AMPO was compared with model predictions. After confirming that measurements closely matched predictions, the model was used to investigate the overall influence of SSC parameters on the maximally attainable AMPO of human *m. quadriceps femoris*.

2.1 Experiment

2.1.1 Participants and ethical approval

Approval of the local ethics committee was obtained before the start of the *in vivo* experiment (VCWE-2016-135). Four male and two female healthy participants (22.8 ± 2.2 years, mean ± SD), who all signed

informed consent, participated in the experiment.

2.1.2 Measurement setup

A kinematically driven knee dynamometer (custom made, Vrije Universiteit, Amsterdam, the Netherlands, see Ruiter et al., 2006) was used to fully impose periodic knee joint angle over time. This dynamometer was capable of imposing large knee joint accelerations and was sufficiently stiff such that participant behaviour (i.e. knee joint moment changes) did not influence the resulting knee joint movement (see Results). During the experiment, participants sat in the dynamometer with the trunk, hip, and thigh firmly strapped. The foot was secured to the dynamometer footplate to fix the ankle joint. The fixed joint and segment angles were as follows: the hip angle (between trunk and thigh, 0 rad = fully extended) was about 1.2 rad, the ankle angle (between shank and foot, 0 rad = fully extended) was about 1.2 rad and the thigh angle (relative to the horizontal) was about 3.3 rad (see Figure 2). The knee joint angle was defined such that 0 rad corresponded to a fully extended knee, while knee joint angle increased with flexion (see Figure 2). Participants performed the experiment with their preferred leg. The lower leg was attached to the sliding carriage of the dynamometer just proximal to the lateral malleolus (see Figure 2). To align the knee joint with the rotation axis of the dynamometer, the lateral epicondyle was initially used as a proxy for the knee joint axis while the participant generated a substantial knee extension moment. Next, the participant actively extended and flexed their knee over the experimental joint angle range. During this movement, the travel of the sliding carriage was visually inspected, and the knee joint alignment was adjusted to minimise carriage displacement. Net knee joint moment and knee joint angle were measured with a torque sensor (custom made, Vrije Universiteit, Amsterdam, the Netherlands) and an angle sensor (custom made, Vrije Universiteit, Amsterdam, the Netherlands) at a sample rate of 1000 Hz.

Surface EMG of the *m. vastus medialis*, taken as the reference muscle for *m. quadriceps femoris*, was measured during the AMPO measurements. In addition, EMG of *m. biceps femoris*, taken as the reference muscle for the hamstring muscle group, was measured to test for undesired co-contraction. Bipolar EMG electrodes (Ambu, Ballerup, Denmark) were placed at an interelectrode distance of 20mm, at 4/5 of the line originating from the anterior spina iliaca superior to the joint space in front of the anterior border of the medial ligament and at 1/2 of the line originating from the ischial tuberosity to the lateral epicondyle of the tibia for *m. vastus medialis* and *m. biceps femoris* respectively (SENIAM guidelines, Hermens et al., 2000). EMG recordings were amplified with a pre-amplifier (9200 EMG, Goodvaerts Instruments, Kortenhoef, the Netherlands) and collected at a sample rate of 4000 Hz.

To synchronise the dynamometer and EMG measurement system, a trigger pulse generated by the dynamometer system was sent to the EMG measurement system at the start of each measurement. In addition, knee joint angle was recorded simultaneously by both systems. After acquisition, the datasets were aligned based on the trigger pulse. To verify correct synchronisation, the knee joint angle signals recorded by the two systems were compared. The signals overlapped after alignment (i.e., maximal cross-correlation occurred at a time shift of 0), confirming successful time synchronisation between the

dynamometer and EMG recordings. The EMG signals were downsampled from 4000 Hz to 1000 Hz, beingthe sampling rate of the dynamometer system. Before downsampling, a low-pass anti-aliasing filter with a cut-off frequency of 500 Hz (dynamometer sample rate divided by 2) was applied to prevent aliasing artifacts. All subsequent analyses were performed on the synchronised and downsampled datasets.

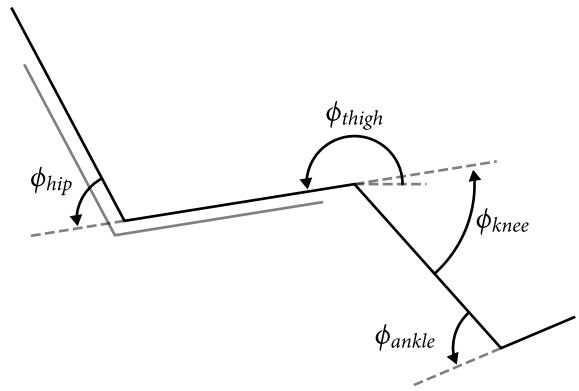

Figure 2: The experimental setup.

2.1.3 Isometric measurements

The maximally attainable AMPO of participants depends on their maximal isometric force. Consequently, for a meaningful comparison between the magnitude of measured AMPO and predicted maximally attainable AMPO, as well as for meaningful comparisons between participants, we needed an accurate estimate of each participant's maximal isometric force. To obtain this estimate, we measured the isometric knee joint moment-angle relationship for each participant individually. Maximal isometric net knee joint moment was determined at nine different knee joint angles equidistantly distributed over a range of knee joint angles between 0.2 rad and 1.8 rad. The order of these measurements was assigned pseudo-randomly such that they were equally distributed across participants. Participants were instructed to maximise their net knee joint moment with their *m. quadriceps femoris* at every knee joint angle for a period of about five seconds. Participants received real-time visual feedback about their net knee joint moment on a monitor which was placed in front of them and were verbally encouraged by the experimenter. There was a 3-minute rest between each isometric measurement. At the end of the experiment, the isometric measurements for the first two knee joint angles were repeated to assess fatigue.

2.1.4 AMPO measurements

Experimental conditions for the AMPO measurements were selected based on simulations using a Hill-type MTC model. Due to limits on knee angular acceleration, only a subset of combinations of cycle frequency, FTS and knee joint excursion were experimentally feasible (for details, see Appendix). Nine experimental conditions that differed in cycle frequency and FTS at a knee joint excursion of 1.6 rad were used in the experiment (see Figure 3). One subset of five experimental conditions was predicted to

yield identical maximally attainable AMPO despite substantial variations in cycle frequency and FTS (0.35A-E; Figure 3). A second subset of five experimental conditions was predicted to yield substantial differences in maximally attainable AMPO due to variations in cycle frequency and FTS (0.20, 0.25, 0.30, 0.35A-E and 0.40; Figure 3). One experimental condition was shared between both subsets, resulting in nine unique experimental conditions.

In each experimental condition, 15 consecutive cycles of the respective knee joint movement were imposed; the participants were instructed to remain fully relaxed during cycles 1-5 and cycles 11-15 and to maximise AMPO with their *m. quadriceps femoris* during cycles 6-10. Throughout all cycles, participants were instructed to relax their hamstring muscle group to prevent it from producing negative and positive mechanical work during knee extension and flexion.

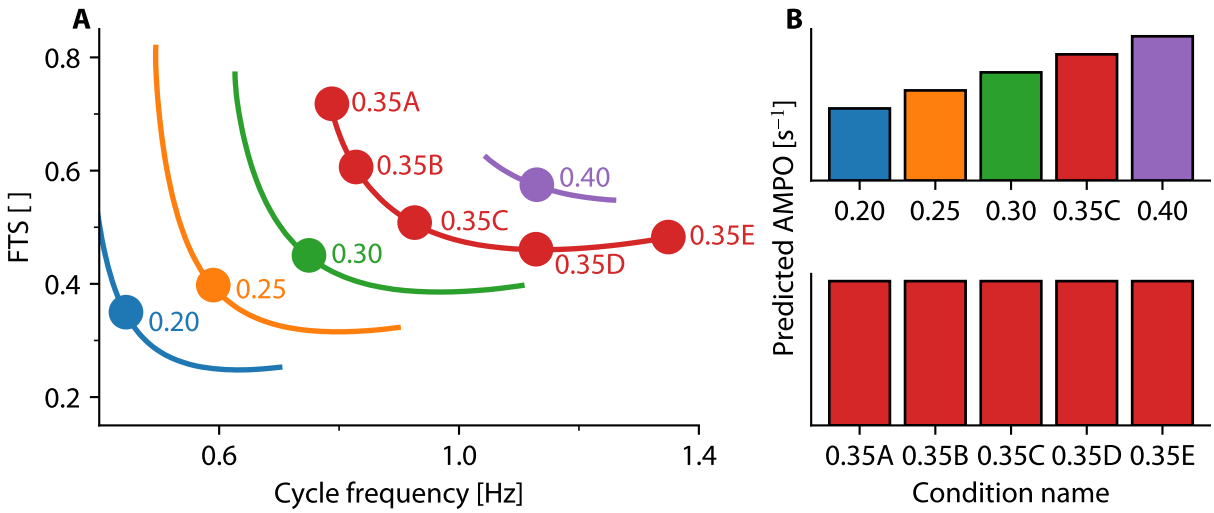

Figure 3: Predicted maximally attainable AMPO and experimental conditions. A) Contour lines depict model-based predictions of maximally attainable AMPO as a function of cycle frequency and FTS for a knee joint excursion of 1.6 rad. Contour lines are only shown for experimentally feasible conditions (due to limits set on knee joint acceleration; see Appendix). Dots indicate the nine conditions selected to be evaluated experimentally. B) Barplots of the same conditions: top, subset predicted to differ in maximally attainable normalised AMPO (i.e. 0.20, 0.25, 0.30, 0.35C and 0.40); bottom, subset predicted to yield identical maximally attainable normalised AMPO (0.35A-E).

The periodic knee joint angle over time was fully imposed by the knee dynamometer and therefore periodic lengthening and shortening of the MTC was fully imposed. Consequently, participants could only influence measured AMPO through their muscle activation. For the predictions derived with the Hill-type MTC model, muscle activation was optimised for AMPO (for details, see Methods: Musculoskeletal model). Thus, for a meaningful comparison between measured AMPO and the predicted maximally attainable AMPO, it was crucial that participants chose a (near-)optimal muscle activation of their *m. quadriceps femoris*. Therefore, each participant practised the nine conditions twice during both practice days (day 1 and 2) of the experiment. Additionally, all conditions were performed once at the end of day 3, at which day the isometric measurements took place. Thus, in total, participants practised

each of the nine conditions five times, equalling 25 active cycles. During both practice and the actual AMPO measurements, participants received real-time visual feedback about their cumulative work throughout each cycle on a monitor and were verbally encouraged by the experimenter to maximise it.

During the AMPO measurements day (day 4), EMG measurements of *m. quadriceps femoris* and the hamstring muscle group were taken. At the start and at the end of the AMPO measurement, participants performed isometric contractions at a knee joint angle of 1.0 rad: twice for *m. quadriceps femoris* and twice for the hamstring muscle group in order to record EMG signals during maximal voluntary contractions (EMG^{MVC}) of both muscles, which were later used to normalise EMG activity during the AMPO measurements (see Methods: Data analysis). Before the actual measurement, the participants again practised with the protocol by performing the four most extreme conditions of the experiment (i.e. 0.20C, 0.35A, 0.35E and 0.40C). To prevent fatigue during the actual measurements, participants were instructed to perform these conditions sub-maximally. After this, participants performed the nine different conditions, from which the first two were repeated at the end to check for muscle fatigue. As with the isometric measurements, the order of the AMPO conditions was assigned pseudo-randomly such that they were equally distributed across participants. There was a 3-minute rest between each condition.

2.2 Musculoskeletal model

2.2.1 Description

A planar model consisting of an upper and lower leg connected by a frictionless hinge joint was used to relate knee joint angle to muscle-tendon-complex (MTC) length and to relate MTC force to knee joint moment (for details, see Appendix). The musculoskeletal model did not contain any knee flexors, as participants were instructed to maximise AMPO using their knee extensors only. A single Hill-type MTC model was used to represent the mechanical behaviour of all knee extensors in the human leg. The formulation of the Hill-type muscle-tendon complex model has been previously described in full detail (Reuvers and Kistemaker, 2025; Soest and Bobbert, 1993). Briefly, the Hill-type MTC model included a contractile element (CE) with a parallel elastic element (PEE) in parallel and a serial elastic element (SEE) in series with the CE. PEE and SEE were modelled as purely passive elastic structures, with their force non-linearly dependent on their length. CE force depended on CE length, CE velocity and active state. Active state, defined as the relative Ca^{2+} concentration bound to troponin C (Ebashi and Endo, 1968), depended on free Ca^{2+} concentration and CE length (Hatze, 1981, pp 31-32; Kistemaker et al., 2005). The free Ca^{2+} concentration in turn, depended on neural drive ($STIM$) by first-order dynamics. $STIM$ and the imposed knee joint angle over time were the model's independent inputs.

2.2.2 Predictions of maximally attainable AMPO

To derive predictions of the maximally attainable AMPO, we used the musculoskeletal model with parameter values obtained from literature (see Table A1). As such, we derived generic predictions without tuning those to the muscle parameters of the participants in our experiment. Using the musculoskeletal model, we imposed periodic knee joint movements in which knee joint angle was

centred around 1.0 rad. We set the knee angular velocities to constant values that could differ between flexion and extension. We examined a large number of knee joint movements over a range of cycle frequencies (0.4-3 Hz, in 0.2 Hz steps), FTS values (0.10-0.90, in 0.05 steps) and knee joint excursions (0.8-2.0 rad, in 0.1 rad steps). Muscle activation onset time (i.e. the time instance at which $STIM$ switches from 0 to its maximal value) was chosen at the start of MTC shortening for each imposed knee joint movement. Muscle activation duration (i.e. the time period during which $STIM$ remains at its maximal value before returning to 0) was optimised for AMPO using a Nelder-Mead Simplex method (Lagarias et al., 1998). To ensure periodic behaviour, simulations were run for at least five cycles and until the difference in MTC force at the beginning and end of the cycle was less than 0.1% F_{CE}^{max}. AMPO was calculated for the last cycle.

2.3 Data analysis

The isometric measurements were used to estimate the maximal isometric force (F_{CE}^{max}) for each participant individually. The maximal net knee joint moment was determined for every measured knee joint angle and was defined as the highest 500 ms average after correcting for gravitational effects. The moment arm was assumed to be equal for all participants (i.e. moment arm = 4.2 cm, see Soest and Casius (2000)). The isometric knee joint moment-angle relationship of the musculoskeletal model depends on three parameters: F_{CE}^{max}, L_{CE}^{opt} (CE optimum length), and L_{SEE}^{0} (SEE slack length). These parameters were estimated individually for each participant by minimising the sum of squared differences between the measured isometric net knee joint moment and those predicted by the musculoskeletal model. This optimisation was performed using a Nelder-Mead Simplex method (Lagarias et al., 1998).

EMG signals measured during the AMPO measurements were bandpass filtered using a bidirectional first order Butterworth filter with cut-off frequencies of 20 and 500 Hz (Butterworth, 1930). The EMG envelope was determined as the absolute value of the Hilbert transformed EMG signal (Hilbert, 1904). EMG^{MVC} values of *m. vastus medialis* and *m. biceps femoris* were defined as the maximal average value of the EMG envelope over 500 ms during the isometric contractions at a knee joint angle of 1.0 rad. Measured EMG signals of *m. vastus medialis* and *m. biceps femoris* during the AMPO measurements were smoothed by a moving average filter with a window of 100 ms and were normalised by their EMG^{MVC} value. The first and last 0.1 seconds of knee extension were not taken into account for the calculation of the average EMG values, as EMG values increased and decreased during these time intervals. Therefore, average EMG values reported were calculated between a knee joint angle of 0.45 and 1.55 rad.

The knee joint moment measured by the dynamometer during the AMPO measurements represented the combined contributions of gravity, inertia (from both the participant's leg+foot and that of the dynamometer), passive structures, and active muscle force. To estimate the contribution of gravity, inertia and passive structures, participants were instructed to remain fully relaxed during the first and last five cycles of each experimental condition. The knee joint moment recorded during these passive cycles was assumed to reflect only the net non-muscular contribution. To isolate the muscular

contribution, the average knee joint moment from passive cycles 2, 3, 4, 12, 13, and 14 was calculated and subtracted from the measured knee joint moment. The resulting moment, attributed to the moment resulting from muscle force only, was called the 'muscular' knee joint moment (M_{mus}). AMPO per individual cycle was calculated by integrating M_{mus} over knee joint angle using a trapezoid method, and dividing the result by the cycle duration.

Measured AMPO and predicted maximally attainable AMPO were normalised in two different ways. First, we assessed whether the magnitude of measured AMPO could be adequately predicted. The predicted maximally attainable AMPO scales linearly with maximal isometric force (F_{CE}^{max}) and also depends on CE optimum length (L_{CE}^{opt}). The sensitivity for L_{CE}^{opt} was, however, low: a 10% reduction in L_{CE}^{opt} resulted in only a 3.7 ± 0.3% decrease in AMPO across all experimental conditions. We used the product of the generic model parameter value of L_{CE}^{opt} (see Table A1) and each participant's individually estimated F_{CE}^{max} to normalise measured AMPO. The predicted maximally attainable AMPO was normalised to the generic model parameter values of L_{CE}^{opt} and F_{CE}^{max}. Second, we evaluated how well differences in AMPO between experimental conditions were predicted. For each subject, the mean (μ) and standard deviation (σ) of measured AMPO across all experimental conditions were calculated, and individual condition values were transformed into z-scores for each condition (denoted by the subscript i):

$$z_i = \frac{AMPO_i - \mu}{\sigma} \tag{1}$$

This procedure removed the influence of all potential errors affecting the magnitude of measured AMPO and isolates the relative changes of AMPO. Model-predicted maximally attainable AMPO values were similarly transformed into z-scores, allowing a direct comparison between the measured and predicted z-scores. Importantly, regardless of the normalisation method, whether by the product of L_{CE}^{opt} and F_{CE}^{max} or by z-scores, measured AMPO was assessed relative to the maximally attainable AMPO calculated from the generic model parameters. Thus, normalised model predictions were based on generic model parameters obtained from literature.

3 Results

3.1 AMPO-scaling factors were successfully obtained

All six participants were able to execute the isometric measurements at all imposed knee joint angles. To test for fatigue, the first two knee joint angles were repeated at the end of the isometric measurements. The maximal knee joint moment was 6.6 ± 5.9% lower for the last two measurements compared to the first two measurements, averaged over all participants and the two repeated measurements. The normalised root mean square difference between measured and predicted isometric net knee joint moments was 5.1 ± 2.4% on average. Variance in the estimated values of L_{CE}^{opt} and L_{SEE}^{o} between participants was low (Table 1). A representative example of the measured and modelled maximal isometric knee joint moment as a function of knee joint angle is depicted in Figure 4.

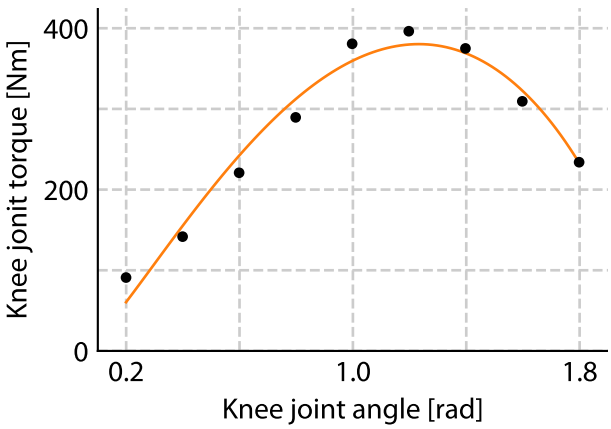

Figure 4: Representative example of maximal isometric knee joint moment as a function of knee joint angle, shown for participant 5. Maximal knee joint moment was measured (black dots) and used to estimate L^{o}_{SEE}, L^{opt}_{CE} and F^{max}_{CE}. The resulting maximal isometric knee joint moment of the musculoskeletal model is depicted by the orange line.

Table 1: Estimated parameter values for each individual participant.

Participant	F^{max}_{CE} [N]	L^{opt}_{CE} [cm]	L^{o}_{SEE} [cm]	**RMSD [%]**
1	4400	8.6	17	7.3
2	5100	9.9	15	6.9
3	5600	8.7	17	5.8
4	6800	8.6	16	4.1
5	9200	7.4	18	4.7
6	8500	8.8	16	2.9
AVG±SD	6600 ± 1700	8.7 ± 0.7	17 ± 1	5.3 ± 1.6

F^{max}_{CE}, L^{opt}_{CE} and L^{o}_{SEE} were estimated from the isometric measurements for each participant individually. The root mean square difference (RMSD) was normalised by the individual maximal predicted knee joint moment, which is the product of F^{max}_{CE} and the moment arm of *m. quadriceps femoris*.

3.2 Knee joint movements were successfully imposed

```
# %% [markdown]
# ---
# title: Analyse imposed knee joint movements
# ---

# %% [markdown]
"""
```

```
The analysis of this page corresponds to the section 'Knee joint movements
were succesfuly imposed'.

Specifically, the following steps were taken:

-   Load condition information (cycle frequency, fraction of time spent shortening) from CSV.
-   Loop over each participant and condition:
    -   Load preprocessed AMPO measurement data (day 4).
    -   Extract relevant columns: time, knee angle (phi), torque, EMG, phase.
    -   Segment the time and angle data into cycles, extension, and flexion phases.
    -   Calculate extension time, flexion time, maximum and minimum angles for each cycle.
    -   Compute the expected theoretical knee angle trajectory for each cycle.
    -   Compute root mean square difference (RMSD) between measured and theoretical trajectories.
-   Compute RMSD statistics across participants, conditions, and cycles.
-   Calculate differences between measured and expected extension/flexion times.
-   Calculate differences in range of motion (ROM) between measured and theoretical angles.

Custom functions used:

-   `helpers.segments_to_list(time, phase, segment_type, cycle_indices)`
    : Segments the continuous time series data into lists of individual cycles, extension, or flexi
-   `trajectories.dino(time_vec, cf, fts, kje, phi_avg, acc, first_contr)`
    : Computes knee angle over time for the dynamometer based on SSC parameters.
-   `stats.rmse(signal1, signal2)`
    : computes root-mean-square error between signals.

"""

# %% [markdown]
"""
## Load packages & set directories
"""

#%% Load packages
import os
import sys
import pandas as pd
import numpy as np
```

```python
from pathlib import Path

# Set directories
cwd = Path.cwd()
baseDir = cwd.parent
dataDir = baseDir / 'data'
funcDir = baseDir / 'analysis' / 'functions'
sys.path.append(str(funcDir))

# Import custom functions for data processing
import helpers, stats, trajectories

# %% [markdown]
"""
## Compute variables for each participant and condition
"""

#%% Compute variables for each participant and condition
# Load conditions data from CSV file
data = pd.read_csv(os.path.join(dataDir, 'conditions.csv'), sep=',', header=None).to_numpy()

# Define condition names
condNames = ['0.40', '0.35C', '0.30', '0.25', '0.20', '0.35A', '0.35B', '0.35D', '0.35E']

# Extract condition data for cycle frequencies (cf) and flexion time (fts)
cf = data[1].astype(float)
fts = data[2].astype(float)

# Calculate the expected extension and flexion times per condition
TextCond = fts / cf  # Extension time
TflxCond = (1 - fts) / cf  # Flexion time

# Define maximum and minimum knee angles for conditions
phiMaxCond = 1.8  # Max knee angle (radians)
phiMinCond = 0.2  # Min knee angle (radians)

# List of participants
pps = [1, 2, 3, 4, 5, 6]
```

```python
# Initialize matrices for storing processed data
Ncycli = 14  # Number of cycles
Text = np.nan * np.zeros((len(pps), len(condNames), Ncycli))
Tflx = np.nan * np.zeros((len(pps), len(condNames), Ncycli))
phiMax = np.nan * np.zeros((len(pps), len(condNames), Ncycli))
phiMin = np.nan * np.zeros((len(pps), len(condNames), Ncycli))
rmsd = np.nan * np.zeros((len(pps), len(condNames), Ncycli))

day = 4  # Day of experiment
trial = 1  # Trial number

for iPP, pp in enumerate(pps):
    # Define directory for each participant's preprocessed data
    dataDirPP = os.path.join(dataDir, 'dataExp', f'pp{pp:02d}', 'AMPO measurements', '')

    for iCond, condName in enumerate(condNames):
        # Define filename for each condition
        filename = f'pp{pp:02d}_{condName}_t1'

        # Load the preprocessed data
        df = pd.read_csv(os.path.join(dataDirPP, filename + '.csv'))
        data = df.to_numpy()

        # Extract columns from the data
        time, phi, _, torqueComp, _, _, emgQuad, emgHams, phase = data.T[:9]

        # Process the extension and flexion times using the custom function
        mTimeExt = helpers.segments_to_list(time, phase, 'ext', np.arange(0, 14))
        mTimeFlx = helpers.segments_to_list(time, phase, 'flx', np.arange(0, 14))

        mTime = helpers.segments_to_list(time, phase, 'cycle', np.arange(0, 14))
        mPhi = helpers.segments_to_list(phi, phase, 'cycle', np.arange(0, 14))
        mPhiExt = helpers.segments_to_list(phi, phase, 'ext', np.arange(0, 14))

        # Calculate extension time, flexion time, max and min angles for each cycle
        for iCycle in range(Ncycli):
            Text[iPP, iCond, iCycle] = mTimeExt[iCycle][-1] - mTimeExt[iCycle][0]
            Tflx[iPP, iCond, iCycle] = mTimeFlx[iCycle][-1] - mTimeFlx[iCycle][0]
```

```
            phiMax[iPP, iCond, iCycle] = mPhiExt[iCycle].max()
            phiMin[iPP, iCond, iCycle] = mPhiExt[iCycle].min()

            phiTheor = trajectories.dino(mTime[iCycle]-mTime[iCycle][0],cf[iCond],fts[iCond],kje=1
            rmsd[iPP, iCond, iCycle] = stats.rmse(mPhi[iCycle],phiTheor)

# Root mean square differences
rmsd_mean = f"{rmsd.mean()*1e3:0.0f}" # [mrad]
rmsd_std  = f"{rmsd.std()*1e3:0.0f}"  # [mrad]

# Output ROM results
print(f"RMSD is: {rmsd_mean} ± {rmsd_std} mrad")

# Flexion and Extension Time Differences
# Create matrices for the condition's expected times
a = np.repeat([TextCond.T], 6, axis=0)  # Repeat for each participant
b = a[:, :, None]
TextCond = np.repeat(b, Ncycli, axis=2)

a = np.repeat([TflxCond.T], 6, axis=0)
b = a[:, :, None]
TflxCond = np.repeat(b, Ncycli, axis=2)

# Calculate the mean and standard deviation of time differences
Text_mean = f"{(Text - TextCond).mean() * 1e3:0.1f}"  # [ms] mean of extension time difference
Text_std = f"{(Text - TextCond).std() * 1e3:0.1f}"  # [ms]  std of extension time difference
Tflx_mean = f"{(Tflx - TflxCond).mean() * 1e3:0.1f}"  # [ms]  mean of flexion time difference
Tflx_std = f"{(Tflx - TflxCond).std() * 1e3:0.1f}"  # [ms] std of flexion time difference

# Output the results
print(f"Difference in extension time (desired - measured) is: {Text_mean} ± {Text_std} ms")
print(f"Difference in flexion time (desired - measured) is: {Tflx_mean} ± {Tflx_std} ms")

# Range of Motion (ROM) Calculation
# Calculate ROM (difference between max and min angles) for each cycle
rom_mean = f"{((phiMax - phiMin) - (phiMaxCond - phiMinCond)).mean() * 1e3:0.1f}" # [mrad] ROM mean
rom_std  = f"{((phiMax - phiMin) - (phiMaxCond - phiMinCond)).std() * 1e3:0.1f}"  # [mrad] ROM std
```

```
# Output ROM results
print(f"Difference in range of motion (desired - measured) is: {rom_mean} ± {rom_std} mrad")
```

For a meaningful comparison between the measured and predicted maximally attainable AMPO, it was crucial that the knee dynamometer adequately imposed the desired knee joint movement of the experimental conditions. This to ensure that both measured and predicted AMPO were based on (near-)identical knee joint movements. Averaged across all participants, conditions and cycles, the sum of squared differences between the desired and measured knee joint angle was only $(7 \pm 12) \cdot 10^{-3}$ rad. As a result, knee extension lasted only 0.6 ± 0.5 ms shorter than desired, and knee flexion lasted 0.6 ± 0.5 ms shorter than desired. In addition, the difference between maximum and minimum knee joint angle in the experiment was $(0.7 \pm 0.2) \cdot 10^{-3}$ rad higher than desired, averaged across all participants, conditions and cycles. Overall, the dynamometer successfully imposed the desired knee joint movements both when participants were passive and when they aimed to maximise AMPO.

3.3 Timing of *m. quadriceps femoris* activation was adequate

```
# %% [markdown]
# ---
# title: EMG and mechanical work
# ---

# %% [markdown]
"""
The analysis of this page corresponds to the sections 'Timing of m. quadriceps
femoris activation was adequate' & Mechanical work was achieved almost
exclusively during knee extension'.

Specifically, the following steps were taken:

-   Define participants and experimental conditions.
-   Load preprocessed AMPO measurement data for each participant, condition, and trial.
-   Segment EMG, torque, and knee angle data into cycles, extension, or flexion phases.
-   Cut EMG and torque data to a specific knee angle range (phi_range) for analysis.
-   Compute mean and standard deviation of quadriceps and hamstring EMG during extension and flexio
-   Compute cross-correlation between quadriceps and hamstring EMG signals during extension.
-   Compute absolute mechanical work (Mmus) during flexion over the specified knee angle range.

Custom functions used:

-   `helpers.segments_to_list(data_array, phase_array, part, cycle_indices)`
```

```
    : Segments time series data (EMG, torque, or angle) into individual cycles or specific movemen
-   `helpers.data_cut(data_list, phi_list, phi_range)`
    : Restricts data to a given range of knee angles and returns the cut data for analysis.

"""

# %% [markdown]
"""
## Load packages & set directories
"""

#%% Load packages
import sys
import numpy as np
import pandas as pd
from pathlib import Path

# Set directories
cwd = Path.cwd()
baseDir = cwd.parent
dataDir = baseDir / 'data'
funcDir = baseDir / 'analysis' / 'functions'
sys.path.append(str(funcDir))

# Import custom functions
import helpers

# %% [markdown]
"""
## Helper function to load and process data
"""

#%% Helper function to load and process data
def process_data_for_condition(pp, cond, part = 'cycle', phi_range=[0.45, 1.55], trial=1, iCycles=
    """
    Process data for a given participant (pp) and condition (cond) across multiple cycles.

    Parameters:
```

```
    -----------
    pp : int
        Participant ID.
    cond : str
        Condition identifier.
    trial : int, optional
        Trial number (default is 1).
    iCycles : list, optional
        List of cycles to process (default is [5, 6, 7, 8, 9]).

    Returns:
    --------
    emgQuadCut : list
        Processed EMG data for quadriceps.
    emgHamsCut : list
        Processed EMG data for hamstrings.
    """
    # Construct file path using pathlib
    dataDirPP = dataDir / 'dataExp' / f'pp{pp:02d}' / 'AMPO measurements'
    filename = f'pp{pp:02d}_{cond}_t{trial}'

    # Load the data
    df = pd.read_csv(dataDirPP / (filename + '.csv'))
    data = df.T.to_numpy()

    # Extract relevant data columns
    time, phi, _, torqueComp, _, _, emgQuad, emgHams, phase = data

    # Adjust cycles for specific conditions
    if pp == 3 and cond == '0.35E':
        iCycles = [6, 7, 8, 9, 10]  # For pp3 and cond 0.35E, adjust the cycle indices

    # Extract phase-specific data for flexion/extension
    mPhi        = helpers.segments_to_list(phi, phase, part, iCycles)
    mEMGquad    = helpers.segments_to_list(emgQuad, phase, part, iCycles)
    mEMGhams    = helpers.segments_to_list(emgHams, phase, part, iCycles)
    mTor        = helpers.segments_to_list(torqueComp, phase, part, iCycles)
```

```
    # Cut the EMG data within the specified range
    emg_quad_cut, _ = helpers.data_cut(mEMGquad, mPhi, phi_range)
    emg_hams_cut, _ = helpers.data_cut(mEMGhams, mPhi, phi_range)
    tor_cut, _      = helpers.data_cut(mTor, mPhi, phi_range)

    return emg_quad_cut, emg_hams_cut, tor_cut

#%% Main loop to process all participants and conditions
def process_all_conditions(part = 'cycle', phi_range=[0.45, 1.55]):
    """
    Process all participants and conditions to calculate data over a certain range of knee angles.
    """
    pps = [1, 2, 3, 4, 5, 6]
    conds = ['0.40', '0.35C', '0.30', '0.25', '0.20', '0.35A', '0.35B', '0.35D', '0.35E']

    # Initialize lists to store the results
    emg_quad_tot = []
    emg_hams_tot = []
    tor_tot = []

    # Process each condition and participant
    for cond in conds:
        for pp in pps:
            emg_quad_cut, emg_hams_cut, tor_cut = process_data_for_condition(pp, cond, part, phi_ra

            # Append the processed EMG data to the total list
            emg_quad_tot.extend(emg_quad_cut)
            emg_hams_tot.extend(emg_hams_cut)
            tor_tot.extend(tor_cut)

    return emg_quad_tot, emg_hams_tot, tor_tot

# %% [markdown]
"""
## EMG during extension between phi 0.45-1.55
"""

#%% EMG during extension between phi 0.45-1.55
```

```
emg_quad_tot, emg_hams_tot = process_all_conditions(part='ext', phi_range=[0.45, 1.55])[0:2]

emg_quad_ext_mean = f"{np.concatenate(emg_quad_tot).astype(float).mean()*1e2:0.1f}"
emg_quad_ext_std  = f"{np.concatenate(emg_quad_tot).astype(float).std()*1e2:0.1f}"
emg_hams_ext_mean = f"{np.concatenate(emg_hams_tot).astype(float).mean()*1e2:0.1f}"
emg_hams_ext_std  = f"{np.concatenate(emg_hams_tot).astype(float).std()*1e2:0.1f}"

# Calculate and print the mean and standard deviation for both muscles
print("Quadriceps EMG during extension (between knee angle 0.45-1.55 rad): " +
      f"{emg_quad_ext_mean} ± {emg_quad_ext_std} %")
print("Hamsrings EMG during extension (between knee angle 0.45-1.55 rad): " +
      f"{emg_hams_ext_mean} ± {emg_hams_ext_std} %")

# %% [markdown]
"""
## Cross-correlationf EMGs during extension between phi 0.2-1.8
"""

#%% Cross-correlationf EMGs during extension between phi 0.2-1.8
# Process all conditions and participants with a different phi_range
emg_quad_tot, emg_hams_tot = process_all_conditions(part = 'ext', phi_range=[0.2, 1.8])[0:2]

# Vectorize cross-correlation calculation using numpy
xcorr = [
    np.corrcoef(np.array(x).astype(float), np.array(y).astype(float))[0, 1]
    for x, y in zip(emg_quad_tot, emg_hams_tot)
]

# nanmean because there is 1 participant 1 condition where EMG was not saved..
xcorr_mean = f"{np.nanmean(xcorr):0.2f}"
xcorr_std  = f"{np.nanstd(xcorr):0.2f}"

# Calculate and print the mean and standard deviation of cross-correlation values
print("Cross-correlation between quad & hams EMG is: " +
      f"{xcorr_mean} ± {xcorr_std} %")

# %% [markdown]
"""
```

```
## EMG during flexion between phi 0.45-1.55
"""

#%% EMG during flexion between phi 0.45-1.55
# Process all conditions and participants
emg_quad_tot, emg_hams_tot = process_all_conditions(part='flx', phi_range=[0.45, 1.55])[0:2]

emg_quad_flx_mean = f"{np.concatenate(emg_quad_tot).astype(float).mean()*1e2:0.1f}"
emg_quad_flx_std  = f"{np.concatenate(emg_quad_tot).astype(float).std()*1e2:0.1f}"
emg_hams_flx_mean = f"{np.concatenate(emg_hams_tot).astype(float).mean()*1e2:0.1f}"
emg_hams_flx_std  = f"{np.concatenate(emg_hams_tot).astype(float).std()*1e2:0.1f}"

# Calculate and print the mean and standard deviation for both muscles
print("Quadriceps EMG during flexion (between knee angle 0.45-1.55 rad): " +
      f"{emg_quad_flx_mean} ± {emg_quad_flx_std} %")
print("Hamsrings EMG during flexion (between knee angle 0.45-1.55 rad): " +
      f"{emg_hams_flx_mean} ± {emg_hams_flx_std} %")

# %% [markdown]
"""
## Absolute Mmus during flexion between phi 0.45-1.55
"""

#%% Absolute Mmus during flexion between phi 0.45-1.55
# Process all conditions and participants
Mmus_tot = process_all_conditions(part='flx', phi_range=[0.45, 1.55])[2]

Mmus_mean = f"{np.abs(np.concatenate(Mmus_tot).astype(float)).mean():0.1f}"
Mmus_std = f"{np.abs(np.concatenate(Mmus_tot).astype(float)).mean():0.1f}"

# Calculate and print the mean and standard deviation for Mmus
print("Mmus flexion (between knee angle 0.45-1.55 rad): " +
      f"{Mmus_mean} ± {Mmus_std} Nm")
```

EMG values during knee extension averaged over all conditions and participants were 77.7 ± 31.9% of EMG^{MVC} for *m. quadriceps femoris* and 16.7 ± 12.2% of EMG^{MVC} for the hamstring muscle group, between a knee joint angle of 0.45 and 1.55 rad. There was substantial interindividual variance in EMG values for both muscles, while variance in EMG values between conditions within each participant was small. EMG values of the hamstring muscle group increased just before the transition from flexion to

extension and followed a similar time course as the EMG of *m. quadriceps femoris* (see Figure 5). This was reflected by a correlation coefficient of 0.76 ± 0.18 between EMG values of *m. quadriceps femoris* and of the hamstring muscle group, suggesting possible crosstalk between the two muscle groups. During flexion, EMG values of the hamstring muscle group were close to zero for all participants and conditions. EMG values of *m. quadriceps femoris* increased just before the transition from flexion to extension and were close to zero around the transition from extension to flexion for all participants and conditions (see Figure 5). Consequently, M_{mus} rose around the transition from flexion to extension and was close to zero around the transition from extension to flexion (Figure 6). Except for occasional delays in the activation onset of *m. quadriceps femoris* during the first active cycle (cycle 6), M_{mus} exhibited little variance between cycles (see also Figure 8). In sum, participants were capable of finding an activation of their *m. quadriceps femoris* that appeared to be (near-)optimal.

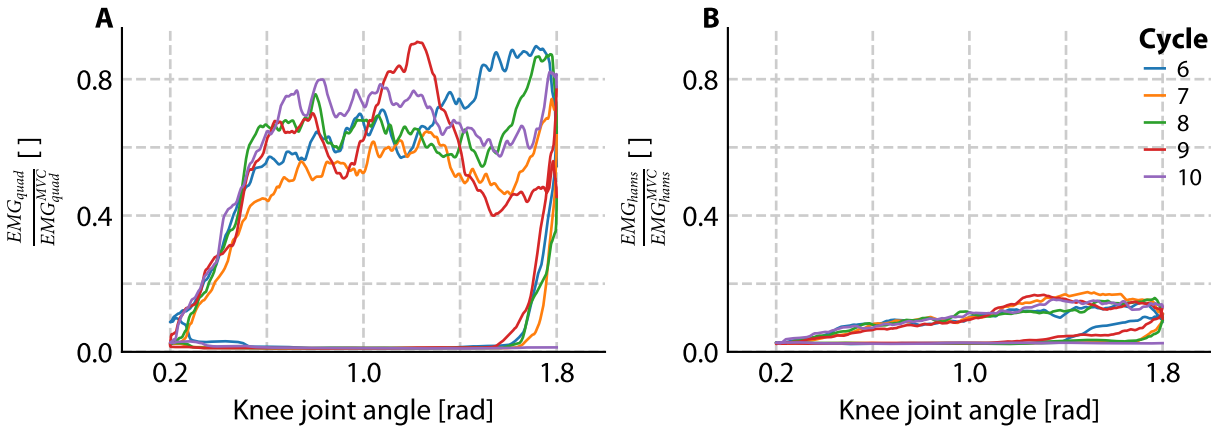

Figure 5: Representative example of measured EMG, shown for participant 5 under condition 0.35C. A) EMG of *m. vastus medialis* normalised by its EMG^{MVC} value as a function of knee joint angle. EMG values increased just before the transition from flexion to extension and were close to zero during flexion, indicating that the timing of *m. quadriceps femoris* activation of participants was adequate. B) EMG of the *m. biceps femoris* normalised by its EMG^{MVC} value as a function of knee joint angle. EMG values increased just before the transition from flexion to extension, following a similar time course as the EMG values of *m. vastus medialis*, suggesting possible co-contraction. During flexion, EMG values were close to zero, indicating participants adequately relaxed their the hamstring muscle group during flexion.

3.4 Mechanical work was achieved almost exclusively during knee extension

```
# %% [markdown]
# ---
# title: EMG and mechanical work
# ---

# %% [markdown]
```

```
"""
The analysis of this page corresponds to the sections 'Timing of m. quadriceps
femoris activation was adequate' & Mechanical work was achieved almost
exclusively during knee extension'.

Specifically, the following steps were taken:

-   Define participants and experimental conditions.
-   Load preprocessed AMPO measurement data for each participant, condition, and trial.
-   Segment EMG, torque, and knee angle data into cycles, extension, or flexion phases.
-   Cut EMG and torque data to a specific knee angle range (phi_range) for analysis.
-   Compute mean and standard deviation of quadriceps and hamstring EMG during extension and flexio
-   Compute cross-correlation between quadriceps and hamstring EMG signals during extension.
-   Compute absolute mechanical work (Mmus) during flexion over the specified knee angle range.

Custom functions used:

-   `helpers.segments_to_list(data_array, phase_array, part, cycle_indices)`
    : Segments time series data (EMG, torque, or angle) into individual cycles or specific movement
-   `helpers.data_cut(data_list, phi_list, phi_range)`
    : Restricts data to a given range of knee angles and returns the cut data for analysis.

"""

# %% [markdown]
"""
## Load packages & set directories
"""

#%% Load packages
import sys
import numpy as np
import pandas as pd
from pathlib import Path

# Set directories
cwd = Path.cwd()
baseDir = cwd.parent
```

```python
dataDir = baseDir / 'data'
funcDir = baseDir / 'analysis' / 'functions'
sys.path.append(str(funcDir))

# Import custom functions
import helpers

# %% [markdown]
"""
## Helper function to load and process data
"""

#%% Helper function to load and process data
def process_data_for_condition(pp, cond, part = 'cycle', phi_range=[0.45, 1.55], trial=1, iCycles=
    """
    Process data for a given participant (pp) and condition (cond) across multiple cycles.

    Parameters:
    -----------
    pp : int
        Participant ID.
    cond : str
        Condition identifier.
    trial : int, optional
        Trial number (default is 1).
    iCycles : list, optional
        List of cycles to process (default is [5, 6, 7, 8, 9]).

    Returns:
    --------
    emgQuadCut : list
        Processed EMG data for quadriceps.
    emgHamsCut : list
        Processed EMG data for hamstrings.
    """
    # Construct file path using pathlib
    dataDirPP = dataDir / 'dataExp' / f'pp{pp:02d}' / 'AMPO measurements'
    filename = f'pp{pp:02d}_{cond}_t{trial}'
```

```python
    # Load the data
    df = pd.read_csv(dataDirPP / (filename + '.csv'))
    data = df.T.to_numpy()

    # Extract relevant data columns
    time, phi, _, torqueComp, _, _, emgQuad, emgHams, phase = data

    # Adjust cycles for specific conditions
    if pp == 3 and cond == '0.35E':
        iCycles = [6, 7, 8, 9, 10]  # For pp3 and cond 0.35E, adjust the cycle indices

    # Extract phase-specific data for flexion/extension
    mPhi        = helpers.segments_to_list(phi, phase, part, iCycles)
    mEMGquad    = helpers.segments_to_list(emgQuad, phase, part, iCycles)
    mEMGhams    = helpers.segments_to_list(emgHams, phase, part, iCycles)
    mTor        = helpers.segments_to_list(torqueComp, phase, part, iCycles)

    # Cut the EMG data within the specified range
    emg_quad_cut, _ = helpers.data_cut(mEMGquad, mPhi, phi_range)
    emg_hams_cut, _ = helpers.data_cut(mEMGhams, mPhi, phi_range)
    tor_cut, _      = helpers.data_cut(mTor, mPhi, phi_range)

    return emg_quad_cut, emg_hams_cut, tor_cut

#%% Main loop to process all participants and conditions
def process_all_conditions(part = 'cycle', phi_range=[0.45, 1.55]):
    """
    Process all participants and conditions to calculate data over a certain range of knee angles.
    """
    pps = [1, 2, 3, 4, 5, 6]
    conds = ['0.40', '0.35C', '0.30', '0.25', '0.20', '0.35A', '0.35B', '0.35D', '0.35E']

    # Initialize lists to store the results
    emg_quad_tot = []
    emg_hams_tot = []
    tor_tot = []

    # Process each condition and participant
```

```
    for cond in conds:
        for pp in pps:
            emg_quad_cut, emg_hams_cut, tor_cut = process_data_for_condition(pp, cond, part, phi_ra

            # Append the processed EMG data to the total list
            emg_quad_tot.extend(emg_quad_cut)
            emg_hams_tot.extend(emg_hams_cut)
            tor_tot.extend(tor_cut)

    return emg_quad_tot, emg_hams_tot, tor_tot

# %% [markdown]
"""
## EMG during extension between phi 0.45-1.55
"""

#%% EMG during extension between phi 0.45-1.55
emg_quad_tot, emg_hams_tot = process_all_conditions(part='ext', phi_range=[0.45, 1.55])[0:2]

emg_quad_ext_mean = f"{np.concatenate(emg_quad_tot).astype(float).mean()*1e2:0.1f}"
emg_quad_ext_std  = f"{np.concatenate(emg_quad_tot).astype(float).std()*1e2:0.1f}"
emg_hams_ext_mean = f"{np.concatenate(emg_hams_tot).astype(float).mean()*1e2:0.1f}"
emg_hams_ext_std  = f"{np.concatenate(emg_hams_tot).astype(float).std()*1e2:0.1f}"

# Calculate and print the mean and standard deviation for both muscles
print("Quadriceps EMG during extension (between knee angle 0.45-1.55 rad): " +
      f"{emg_quad_ext_mean} ± {emg_quad_ext_std} %")
print("Hamsrings EMG during extension (between knee angle 0.45-1.55 rad): " +
      f"{emg_hams_ext_mean} ± {emg_hams_ext_std} %")

# %% [markdown]
"""
## Cross-correlationf EMGs during extension between phi 0.2-1.8
"""

#%% Cross-correlationf EMGs during extension between phi 0.2-1.8
# Process all conditions and participants with a different phi_range
emg_quad_tot, emg_hams_tot = process_all_conditions(part = 'ext', phi_range=[0.2, 1.8])[0:2]
```

```python
# Vectorize cross-correlation calculation using numpy
xcorr = [
    np.corrcoef(np.array(x).astype(float), np.array(y).astype(float))[0, 1]
    for x, y in zip(emg_quad_tot, emg_hams_tot)
]

# nanmean because there is 1 participant 1 condition where EMG was not saved..
xcorr_mean = f"{np.nanmean(xcorr):0.2f}"
xcorr_std  = f"{np.nanstd(xcorr):0.2f}"

# Calculate and print the mean and standard deviation of cross-correlation values
print("Cross-correlation between quad & hams EMG is: " +
      f"{xcorr_mean} ± {xcorr_std} %")

# %% [markdown]
"""
## EMG during flexion between phi 0.45-1.55
"""

#%% EMG during flexion between phi 0.45-1.55
# Process all conditions and participants
emg_quad_tot, emg_hams_tot = process_all_conditions(part='flx', phi_range=[0.45, 1.55])[0:2]

emg_quad_flx_mean = f"{np.concatenate(emg_quad_tot).astype(float).mean()*1e2:0.1f}"
emg_quad_flx_std  = f"{np.concatenate(emg_quad_tot).astype(float).std()*1e2:0.1f}"
emg_hams_flx_mean = f"{np.concatenate(emg_hams_tot).astype(float).mean()*1e2:0.1f}"
emg_hams_flx_std  = f"{np.concatenate(emg_hams_tot).astype(float).std()*1e2:0.1f}"

# Calculate and print the mean and standard deviation for both muscles
print("Quadriceps EMG during flexion (between knee angle 0.45-1.55 rad): " +
      f"{emg_quad_flx_mean} ± {emg_quad_flx_std} %")
print("Hamsrings EMG during flexion (between knee angle 0.45-1.55 rad): " +
      f"{emg_hams_flx_mean} ± {emg_hams_flx_std} %")

# %% [markdown]
"""
## Absolute Mmus during flexion between phi 0.45-1.55
"""
```

```python
#%% Absolute Mmus during flexion between phi 0.45-1.55
# Process all conditions and participants
Mmus_tot = process_all_conditions(part='flx', phi_range=[0.45, 1.55])[2]

Mmus_mean = f"{np.abs(np.concatenate(Mmus_tot).astype(float)).mean():0.1f}"
Mmus_std = f"{np.abs(np.concatenate(Mmus_tot).astype(float)).mean():0.1f}"

# Calculate and print the mean and standard deviation for Mmus
print("Mmus flexion (between knee angle 0.45-1.55 rad): " +
      f"{Mmus_mean} ± {Mmus_std} Nm")
```

Participants were instructed to maximise mechanical work per cycle using their *m. quadriceps femoris*, while keeping their hamstring muscle group completely relaxed at all times. Average net knee joint moment during the passive cycles was subtracted from net knee joint moment during the active cycles to correct for knee joint moment due to inertia, gravity and passive structures (see Figure 6). The resulting knee joint moment (M_{mus}) was assumed to be predominantly the result of muscle force and was, as expected, substantially higher than zero during knee extension for all participants and conditions. Due to activation and deactivation delays, M_{mus} may be above zero just before the transition from flexion to extension and immediately after the transition from extension to flexion, in order to maximise AMPO (see also Figure 8). Yet, M_{mus} should be ideally zero during the middle part of knee flexion. For the active cycle the absolute value of M_{mus} was 3.0 ± 3.0 Nm between a knee joint angle of 0.45 and 1.55 rad (corresponding to 100 ms after and before the transitions). This minimal M_{mus} was a result of low activity of both *m. quadriceps femoris* (2.8 ± 2.7% of EMG^{MVC}) and hamstring muscle group (4.1 ± 4.0% of EMG^{MVC}) during the same time interval. We therefore concluded that participants effectively relaxed their muscles during knee flexion, such that the net mechanical work was almost exclusively the result of positive mechanical work delivered during knee extension (see Table 2).

Table 2: Positive and negative mechanical work during knee extension and flexion as a percentage of the net mechanical work per cycle.

	Condition								
	0.20	**0.25**	**0.30**	**0.35C**	**0.40**	**0.35A**	**0.35B**	**0.35D**	**0.35E**
W_{ext}^{+}	102.3 ± 2.5	102.2 ± 2.0	103.4 ± 3.2	103.3 ± 4.3	104.2 ± 3.5	104.5 ± 4.0	102.5 ± 3.6	101.9 ± 4.2	100.5 ± 2.7
W_{flex}^{+}	1.3 ± 1.2	0.9 ± 0.9	1.1 ± 1.2	0.9 ± 0.7	1.4 ± 1.0	1.6 ± 1.5	1.7 ± 1.2	2.6 ± 1.9	3.3 ± 1.8
W_{ext}^{-}	-0.0 ± 0.1	-0.0 ± 0.0	-0.0 ± 0.0	-0.0 ± 0.0	-0.0 ± 0.0	-0.0 ± 0.0	-0.0 ± 0.0	-0.0 ± 0.0	-0.1 ± 0.4
W_{flex}^{-}	-3.7 ± 2.2	-3.1 ± 2.0	-4.6 ± 3.3	-4.3 ± 4.1	-5.6 ± 3.4	-6.0 ± 3.6	-4.3 ± 3.0	-4.7 ± 4.4	-3.8 ± 2.0

Positive (+) and negative (-) mechanical work during knee extension (ext) and flexion (flex) were calculated by integrating the instantaneous mechanical power over time, separately for the periods when the instantaneous power was positive (positive work) and negative (negative work) respectively.

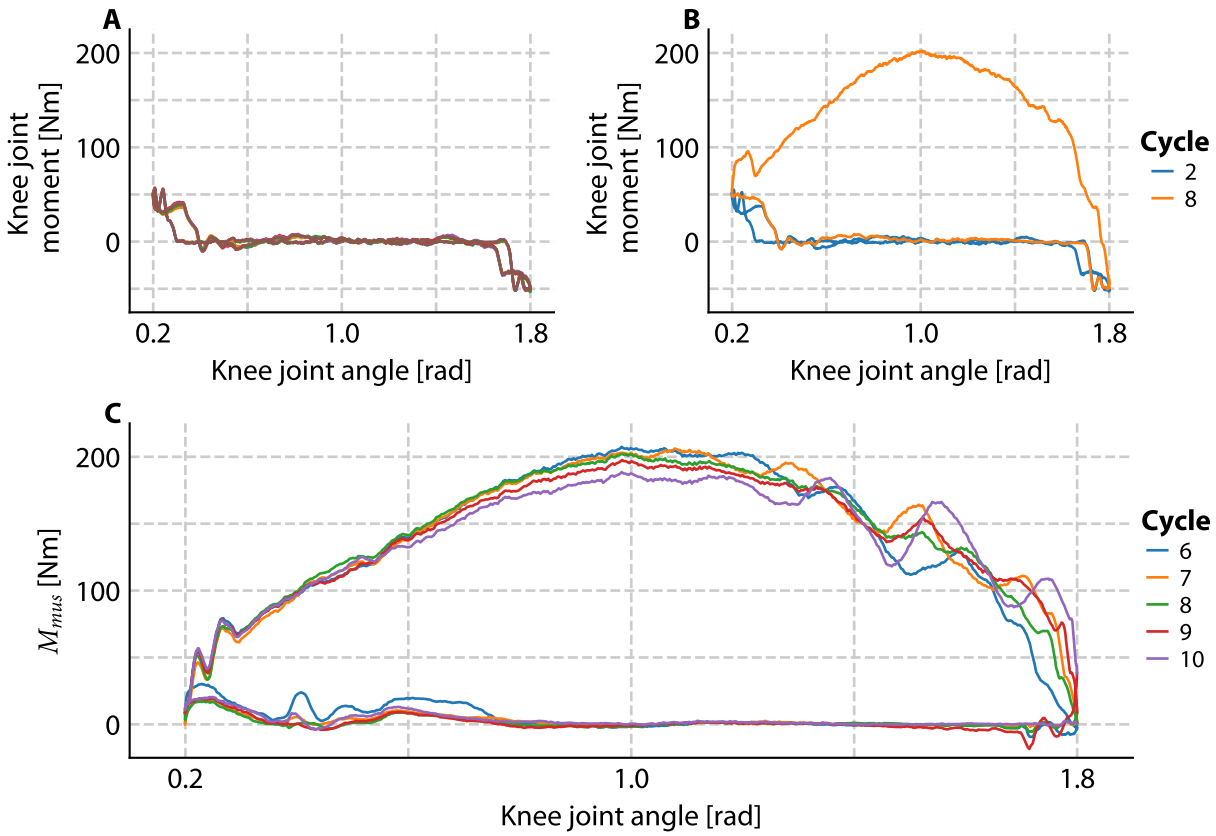

Figure 6: Representative example of mechanical behaviour, shown for participant 5 under condition 0.35C. A) Knee joint moment during passive cycles. Every individual cycle is represented by a different coloured line. B) Knee joint moment during (passive) cycle 2 (blue line) and during (active) cycle 8 (orange line). C) 'Muscular' knee joint moment (M_{mus}) as a function of knee joint angle. M_{mus} was calculated by subtracting average knee joint moment during the passive cycles from the knee joint moment during the active cycles (for details Methods). Every individual cycle is represented by a different coloured line. M_{mus} was close to zero during flexion, indicating that the net mechanical work was almost exclusively the result of mechanical work during extension.

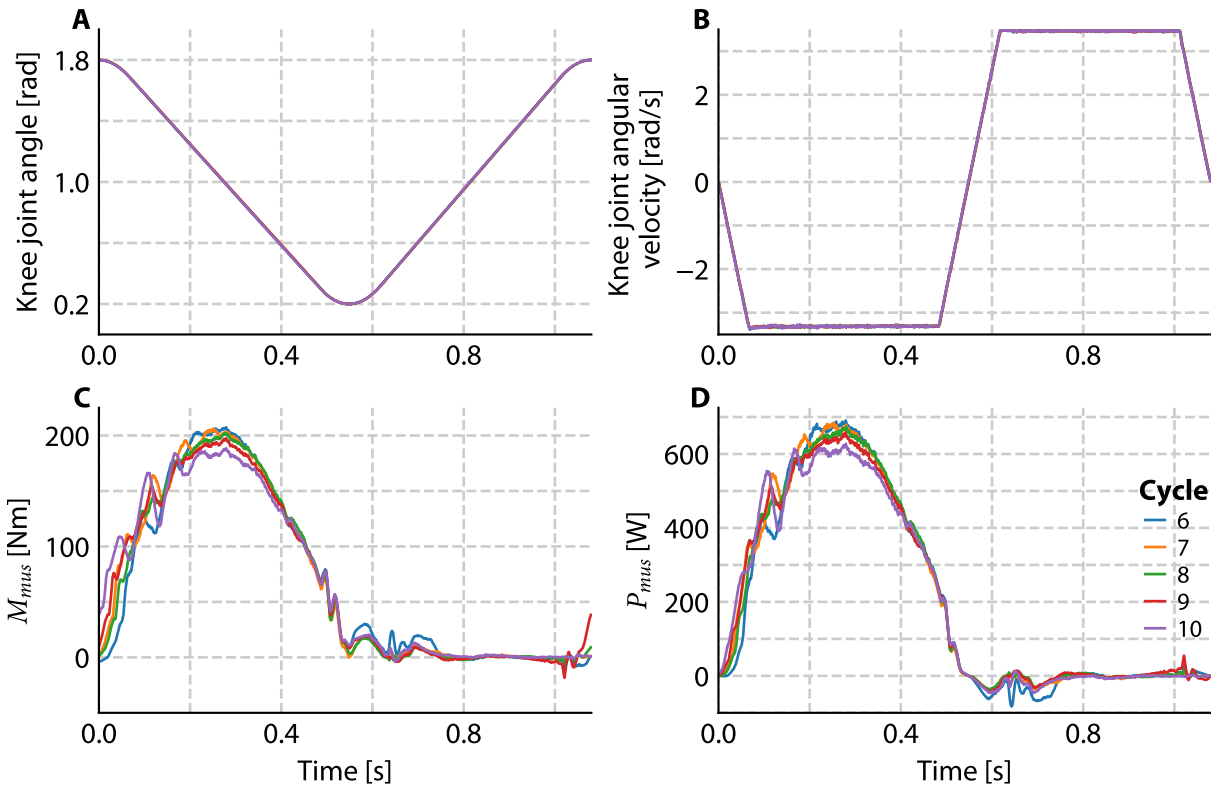

Figure 7: Representative example of mechanical behaviour, shown for participant 5 under condition 0.35C. Time series are only depicted for the active cycles, where time is set to zero at the start of knee extension for each cycle. Individual cycles are represented by a different coloured lines. A) Measured knee joint angle over time (note that these are indistinguishable between cycles). B) Measured knee joint angular velocity over time (indistinguishable between cycles). C) 'Muscular' knee joint moment (M_{mus}) over time, for cycles 6-10 (see Methods). M_{mus} was calculated by subtracting average knee joint moment during the passive cycles from the knee joint moment during the active cycles (see Methods). D) Instantaneous 'muscular' mechanical power (P_{mus}) over time, for cycles 6-10 (see Methods). P_{mus} was calculated as the product of M_{mus} and the measured knee joint angular velocity. Both M_{mus} and M_{Pus} were close to zero during flexion, indicating that the net mechanical work was almost exclusively the result of mechanical work during extension.

3.5 Measured AMPO closely matched predicted maximally attainable AMPO

```
# %% [markdown]
# ---
# title: Measurements against predictions
# format:
#   html:
#       css: styles.css
# ---

# %% [markdown]
"""
This script computes the results of the section 'Measured AMPO closely matched
predicted maximally attainable AMPO'.

For this, it computes:

-   Percent difference between the highest and second highest AMPO within one condition
-   Percent difference between the maximum AMPO of repeated conditions
-   Measured AMPO per participant and condition
-   Standardized z-scores of predicted and measured AMPO
-   Linear regression between predicted and measured AMPO (both normalised and z-scored)
-   Squared Pearson correlation coefficient (r²) for both normalised and z-scored AMPO

Custom functions used:

-   `kinetics.compute_work(phi, torque, phase, part, iCycles)`
    : Computes net, positive, and negative mechanical work per cycle or per phase
-   `stats.pdiff(value1, value2)`
    : Computes percent difference between two values
-   `stats.zscore(array, axis)`
    : Standardizes array to z-score along the specified axis
-   `kinetics.compute_work(phi, torque, phase, part, iCycles)`
    : Computes net, positive, and negative mechanical work per cycle or per phase
"""

# %% [markdown]
"""
## Load packages & set directories
```

```
"""

#%% Load packages
import os, sys
import pandas as pd
import numpy as np
from scipy import stats as sp_stats
from pathlib import Path

# Set directories
cwd = Path.cwd()
baseDir = cwd.parent
dataDir = baseDir / 'data'
funcDir = baseDir / 'analysis' / 'functions'
sys.path.append(str(funcDir))

import kinetics, stats

# %% [markdown]
"""
## Compute percent difference between the highest and second highest AMPO within one condition
"""

#%% Highest vs. second highest AMPO
p_max_vs_2nd = []
ampoDiff = np.nan*np.zeros((6,9))
for iCond, cond in enumerate(['0.40', '0.35C', '0.30', '0.25', '0.20', '0.35A', '0.35B', '0.35D',
    for iPP,pp in enumerate([1,2,3,4,5,6]):
        filepath = os.path.join(dataDir,'dataExp',f'pp{pp:02d}','AMPO measurements',f'pp{pp:02d}_{c
        df = pd.read_csv(filepath)
        data = df.T.to_numpy()

        time,phi,_,torqueComp,_,_,emgQuad,emgHams,phase = data

        iCycles = [5,6,7,8,9]
        if pp == 3 and cond == '0.35E':
            iCycles = [6,7,8,9,10] # for this condition pp3 started one cycle too late!
        Wnet,Wpos,Wneg = kinetics.compute_work(phi,torqueComp,phase,'c',iCycles)
```

```
        WnetSort = np.sort(Wnet)
        p_max_vs_2nd.append(stats.pdiff(WnetSort[-2],WnetSort[-1]))

p_max_vs_2nd = np.array(p_max_vs_2nd) # [%] difference between max. and 2nd max
p_max_vs_2nd_mean = f"{p_max_vs_2nd.mean():0.1f}"  # [%]
p_max_vs_2nd_std  = f"{p_max_vs_2nd.std():0.1f}"  # [%]

print("Difference of highest vs. second highest AMPO within one condition is: " +
      f"{p_max_vs_2nd_mean} ± {p_max_vs_2nd_std} %")

# %% [markdown]
"""
## Compute percent difference between the maximum AMPO of repeated conditions
"""

#%% Difference between max AMPO of regular and repeated condition
repConds = np.array([['0.40', '0.35C'],
                    ['0.35C', '0.30'],
                    ['0.30', '0.25'],
                    ['0.25', '0.20'],
                    ['0.20', '0.35A'],
                    ['0.35A', '0.35B']])

p_rep = []
iCycles = [5,6,7,8,9]
for iPP,pp in enumerate([1,2,3,4,5,6]):
    for iRep in [0,1]:
        repCond = repConds[iPP,iRep]

        filepath = os.path.join(dataDir,'dataExp',f'pp{pp:02d}','AMPO measurements',f'pp{pp:02d}_{
        df = pd.read_csv(filepath)
        data = df.T.to_numpy()
        time,phi,_,torqueComp,_,_,emgQuad,emgHams,phase = data
        Wnet = kinetics.compute_work(phi,torqueComp,phase,'c',iCycles)[0]

        filepath = os.path.join(dataDir,'dataExp',f'pp{pp:02d}','AMPO measurements',f'pp{pp:02d}_{
        df = pd.read_csv(filepath)
        data = df.T.to_numpy()
```

```
        time,phi,_,torqueComp,_,_,emgQuad,emgHams,phase = data
        Wnet_rep = kinetics.compute_work(phi,torqueComp,phase,'c',iCycles)[0]

        p_rep.append(stats.pdiff(Wnet_rep.max(),Wnet.max()))

p_rep = np.array(p_rep) # [%] difference between max. and 2nd max
p_rep_mean = f"{p_rep.mean():0.1f}"  # [%]
p_rep_std  = f"{p_rep.std():0.1f}"  # [%]

print("Difference in highest AMPO of repeated conditions is: " +
      f"{p_rep_mean} ± {p_rep_std} %")

# %% [markdown]
"""
## Compare measured AMPO against predicted maximally attainable AMPO
"""

# %% Measured vs. predicted AMPO
# Load conditions
data = pd.read_csv(dataDir / 'conditions.csv', sep=',', header=None, dtype=str).to_numpy()

# Set conditions order
cond_names = ['0.20', '0.25', '0.30', '0.35C', '0.40', '0.35A', '0.35B', '0.35D', '0.35E']

# Previously I had a different order, so we need to re-order first
current_order = data[0, :]  # e.g., ['0.25', '0.20', '0.35C', ...]
# Find indices that would reorder to match condNames
reorder_idx = [np.where(current_order == c)[0][0] for c in cond_names]
# Reorder all columns
data = data[:, reorder_idx]

c_cf = data[1].astype(float)
c_fts = data[2].astype(float)
c_ampo = data[3].astype(float)

# Participant info
day, trial = 4, 1
pps = [1, 2, 3, 4, 5, 6]
```

```
Npp = len(pps)  # number of participants
filepath = os.path.join(dataDir,'isometric-measurements.csv')
df = pd.read_csv(filepath, index_col=0)

pps = [1,2,3,4,5,6] # participant no.
n_pp = len(pps) # amount of participants
fmax = df['fmax [N]'].to_numpy() # [N] est maximal isometric CE force per particiapnt

# Predictions
ampo_pred = np.array([0.2, 0.25, 0.3, 0.35, 0.4, 0.35, 0.35, 0.35, 0.35])
ampo_pred = np.repeat(ampo_pred[None, :], Npp, axis=0)

# Compute measured AMPO
ampo_exp = np.full((Npp, len(cond_names)), np.nan)
for i_pp, pp in enumerate(range(1, Npp + 1)):
    for i_cond, cond in enumerate(cond_names):
        data_folder = dataDir / "dataExp" / f"pp{pp:02d}" / "AMPO measurements"
        filename = f"pp{pp:02d}_{cond}_t{trial}"

        df = pd.read_csv(data_folder / f"{filename}.csv")
        data = df.T.to_numpy()
        _, phi, _, torque_comp, *_ , phase = data[0:9]

        # Default cycles
        i_cycles = [5, 6, 7, 8, 9]
        if pp == 3 and cond == "0.35E":
            i_cycles = [6, 7, 8, 9, 10]

        # Compute work
        Wnet, Wpos, Wneg = kinetics.compute_work(phi, torque_comp, phase, part="cycle", iCycles=i_c

        # Normalize to participant's max force and convert to AMPO
        ampo_exp[i_pp, i_cond] = np.max(Wnet * c_cf[i_cond]) / (fmax[i_pp] * 0.093)

# Compute mean and SEM across participants
ampo_exp_mean = np.mean(ampo_exp, axis=0)
ampo_exp_sem = np.std(ampo_exp, axis=0, ddof=1) / np.sqrt(Npp)
```

```python
# Standardize for z-score
ampo_pred_z = stats.zscore(ampo_exp, axis=-1)
ampo_exp_z = stats.zscore(ampo_pred, axis=-1)

# Linear regressions
fit_norm = sp_stats.linregress(np.ravel(ampo_pred), np.ravel(ampo_exp))
fit_zsco = sp_stats.linregress(np.ravel(ampo_pred_z), np.ravel(ampo_exp_z))

r2_norm = f"{fit_norm.rvalue ** 2:0.2f}"
r2_zsco = f"{fit_zsco.rvalue ** 2:0.2f}"

print(f"r² (norm AMPO): {r2_norm}")
print(f"r² (z-scored AMPO): {r2_zsco}")

# Measured vs. predicted - absolute values (used in discussion)
p_pred_vs_exp_mean = f"{100 + stats.pdiff(ampo_exp,ampo_pred).mean():0.0f}"
p_pred_vs_exp_std  = f"{stats.pdiff(ampo_exp,ampo_pred).std():0.0f}"

print(f"Measued AMPO is {p_pred_vs_exp_mean} ± {p_pred_vs_exp_std} % " +
      "of predicted AMPO")
```

To assess the reliability of AMPO measurements, we examined the variability of measured AMPO across cycles and repeated conditions. Averaged across each condition and for each participant, the second-highest measured AMPO was only -2.4 ± 2.3% lower than the highest value, indicating low variance in AMPO between cycles. Furthermore, in repeated conditions, the highest measured AMPO at the end of the experiment was just -2.3 ± 4.1% lower than at the start. Together, these results indicate that AMPO measurements were consistent across cycles and throughout the experimental session.

Measured AMPO was compared against the predicted maximally attainable AMPO. First, AMPO was normalised by the generic value of L_{CE}^{opt} and participants' individually estimated F_{CE}^{max} (see Methods: Data analysis). As predicted, (1) measured AMPO differed substantially between conditions 0.20, 0.25, 0.30, 0.35 and 0.40 due to the imposed variations in cycle frequency and FTS and (2) measured AMPO was almost identical for conditions 0.35A-E, despite substantial variations in cycle frequency and FTS (Figure 9; Table 3). Overall, measured AMPO normalised by L_{CE}^{opt} and participants' individual F_{CE}^{max} correlated strongly with predicted maximally attainable AMPO (r^2=0.75). Second, AMPO was normalised by z-scores of measured and predicted maximally attainable AMPO. This revealed a very strong correlation between predictions and measurements (r^2=0.95). All in all, measured AMPO was remarkably close to the predicted maximally attainable AMPO.

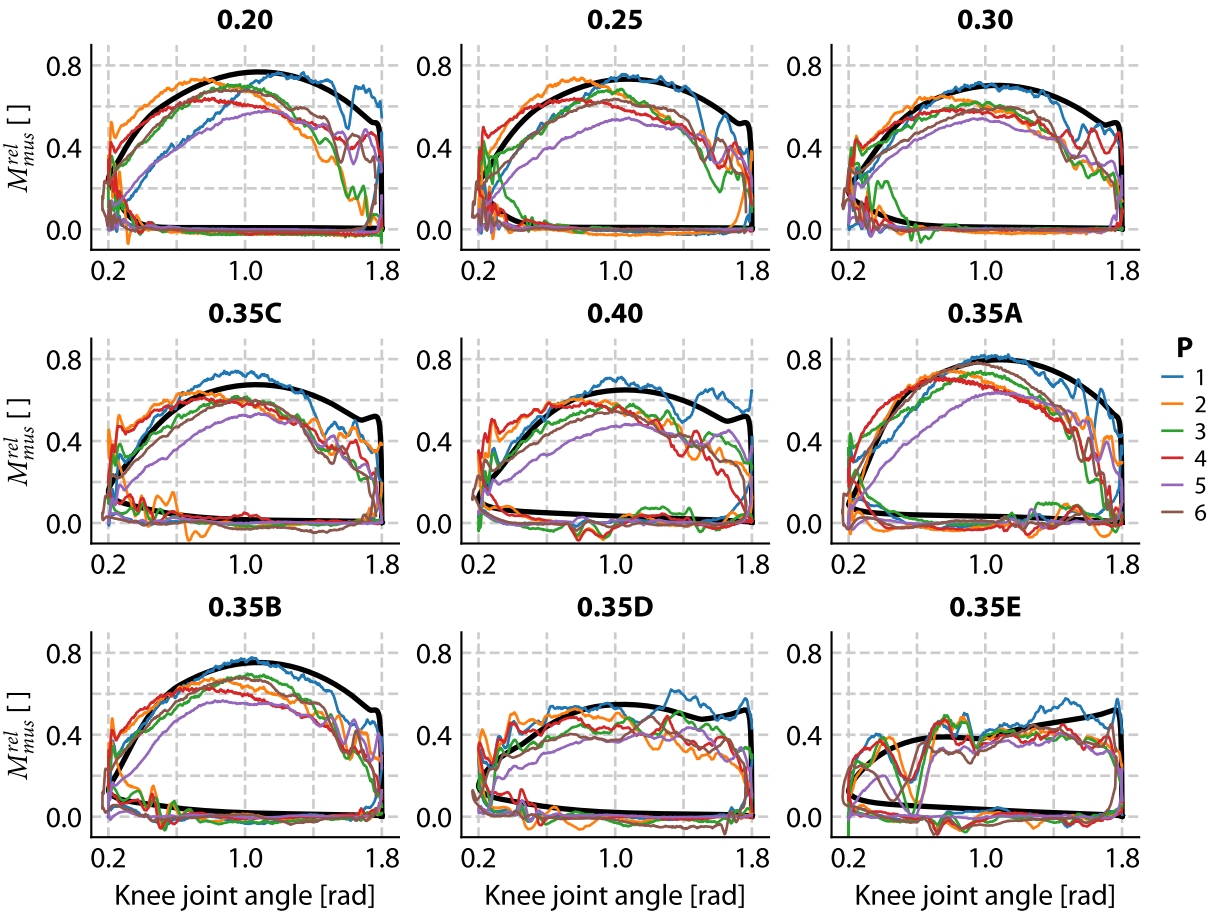

Figure 8: Experimentally estimated and model-predicted relative 'muscular' knee joint moment (M_{mus}^{rel}) for all conditions. M_{mus}^{rel} was calculated by dividing the 'muscular' knee joint moment by the maximal isometric knee joint moment of the individual participants (estimated from the isometric knee joint moment-angle relationship; see Figure 4) or by that of the model. The thick black line depicts the predicted M_{mus}^{rel}, while the thinner coloured lines depicts M_{mus}^{rel} for each individual participant (P). For each participant, M_{mus}^{rel} is shown for the cycle with highest measured AMPO.

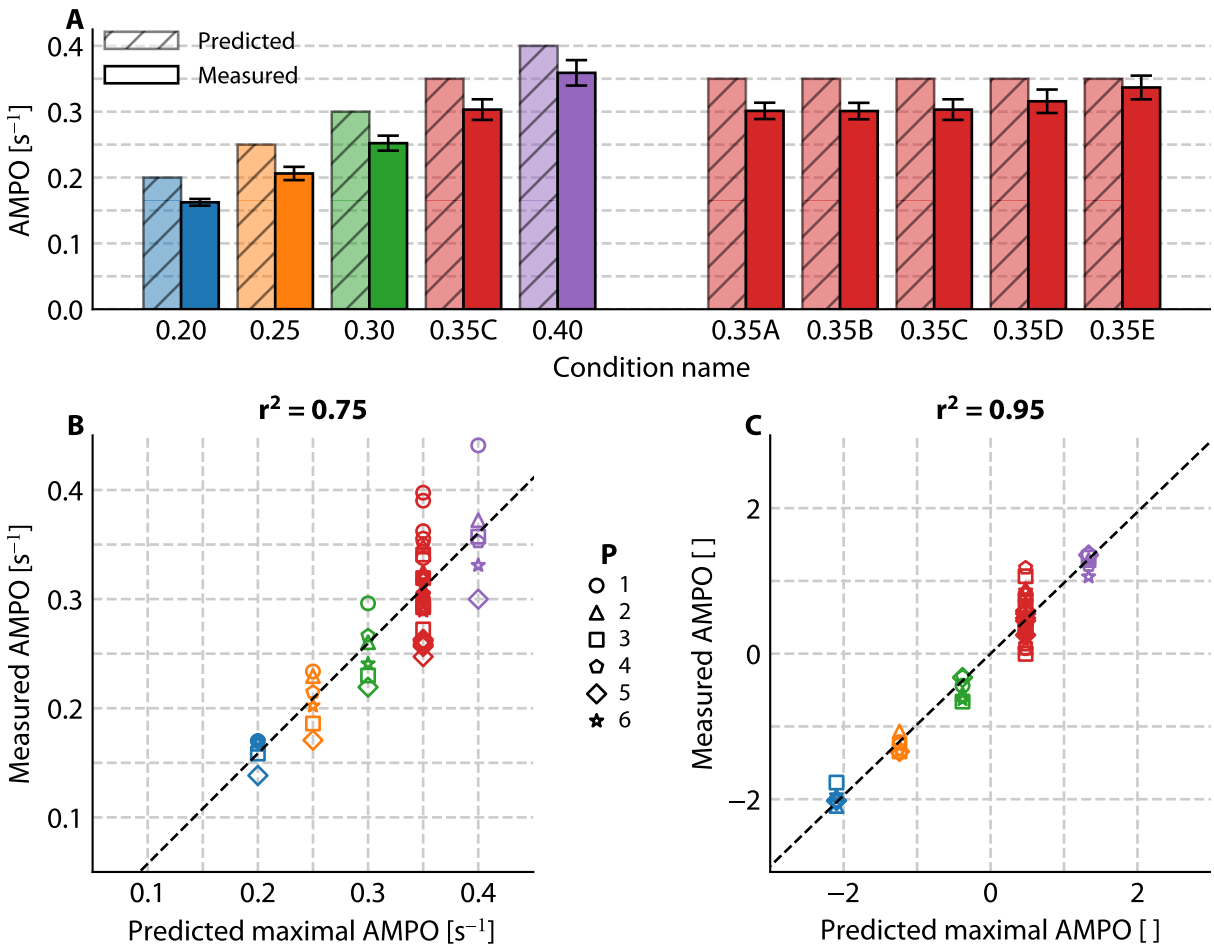

Figure 9: Measured AMPO versus predicted maximally attainable AMPO. A) Barplot showing predicted values with diagonal hatching and semi-transparent fill (α = 0.5), and experimentally measured values fully filled. Error bars denote the standard error of the mean. B) Scatterplot of measured versus predicted maximally attainable AMPO normalised by L_{CE}^{opt} and F_{CE}^{max}, with F_{CE}^{max} estimated individually for each participant. C) Scatterplot of measured versus predicted maximally attainable AMPO normalised to z-scores. Participants (P) are distinguished by symbols.

Table 3: Overview of experimental conditions and measured AMPO per participant and condition.

	Condition								
	0.20	**0.25**	**0.30**	**0.35C**	**0.40**	**0.35A**	**0.35B**	**0.35D**	**0.35E**
									Movement characteristics
CF [Hz]	0.44	0.59	0.75	0.93	1.13	0.79	0.83	1.13	1.35
FTS [ ]	0.35	0.40	0.45	0.51	0.58	0.72	0.61	0.46	0.48
T_{ext} **[s]**	0.79	0.67	0.60	0.55	0.51	0.91	0.73	0.41	0.36
T_{flex} **[s]**	1.46	1.02	0.73	0.53	0.38	0.36	0.48	0.48	0.38
									AMPO
Participant 1 [s⁻¹]	0.17	0.23	0.30	0.36	0.44	0.34	0.36	0.39	0.40
Participant 2 [s⁻¹]	0.17	0.23	0.26	0.32	0.37	0.32	0.30	0.32	0.35
Participant 3 [s⁻¹]	0.16	0.19	0.23	0.29	0.36	0.27	0.29	0.32	0.34
Participant 4 [s⁻¹]	0.17	0.21	0.27	0.30	0.35	0.30	0.29	0.30	0.35
Participant 5 [s⁻¹]	0.14	0.17	0.22	0.25	0.30	0.26	0.26	0.26	0.26
Participant 6 [s⁻¹]	0.17	0.20	0.24	0.29	0.33	0.31	0.30	0.30	0.32
Average ± SD [s⁻¹]	0.16 ± 0.01	0.21 ± 0.02	0.25 ± 0.03	0.30 ± 0.03	0.36 ± 0.04	0.30 ± 0.03	0.30 ± 0.03	0.32 ± 0.04	0.34 ± 0.04
Predicted [s⁻¹]	0.20	0.25	0.30	0.35	0.40	0.35	0.35	0.35	0.35

CF denotes cycle frequency, FTS denotes fraction of cycle time spent shortening, T_{ext} and T_{flex} denote the knee joint extension and flexion time per cycle. Measured AMPO was normalised to the product of the participant's individually estimated F_{CE}^{max} and the generic model parameter value of L_{CE}^{opt}, while predicted AMPO was normalised to the product of model parameters F_{CE}^{max} and L_{CE}^{opt} (see Methods).

3.6 Influence of SSC parameters on the maximally attainable AMPO

```
# %% [markdown]
# ---
```

```
# title: Influence SSC parameters
# ---

# %% [markdown]
"""
This script computes the results of the section 'Influence of SSC parameters
on the maximally attainable AMPO'.

Specifically, it performs the following steps:

-   Loads muscle parameters from a pickle file (`VAS_muspar.pkl`).
-   Defines parameter grids for SSC parameters:
    -   CF (cycle frequency)
    -   FTS (fraction of the cycle time spent shortening)
    -   KJE (knee joint excursion)
-   Loads precomputed simulation data for each combination of SSC parameters.
-   Computes the AMPO (average mechanical power output) normalised by
    maximal isometric CE force (fmax) and optimal CE length (lceopt).
-   Interpolates the AMPO values onto a finer 3D grid for smoother analysis.
-   Determines the maximum AMPO and the corresponding optimal SSC parameters.

"""

# %% [markdown]
"""
## Load packages & set directories
"""

#%% Imports etc.
import os, sys, pickle
import numpy as np
import pandas as pd
from scipy import integrate, interpolate
from pathlib import Path

# Paths
cwd = Path.cwd()
baseDir = cwd.parent
```

```
dataDir = baseDir / 'data'
funcDir = baseDir / 'analysis' / 'functions'
sys.path.append(str(funcDir))

# %% [markdown]
"""
## Extract AMPO of each condition
"""

#%% Extract AMPO of each condition
# Load Parameters
parFile = os.path.join(dataDir, 'VAS_muspar.pkl')
muspar = pickle.load(open(parFile, 'rb'))

# Define parameter grids
kjeSet = np.arange(0.8, 2.01, 0.1)
cfSet = np.arange(0.4, 3.1, 0.2)
ftsSet = np.arange(0.05, 0.96, 0.05)

# Initialize AMPO array
AMPO = np.full((len(kjeSet), len(ftsSet), len(cfSet)), np.nan)

# Load simulation data and compute AMPO
for iKJE, kje in enumerate(kjeSet):
    for iFTS, fts in enumerate(ftsSet):
        for iCF, cf in enumerate(cfSet):
            filepath = os.path.join(dataDir, 'simsSSC',
                                    f'cf{cf:0.2f}Hz_fts{fts:0.2f}_kje{kje:0.2f}rad.csv')
            try:
                df = pd.read_csv(filepath)
                data = df.T.to_numpy()
                time, phi, stim, fsee, gamma, lcerel, q, lmtc, fisomrel = data
                # Compute AMPO normalized by fmax and lce_opt
                AMPO[iKJE, iFTS, iCF] = -integrate.trapezoid(fsee * muspar['A1'], phi) * cf / (musp
            except Exception:
                AMPO[iKJE, iFTS, iCF] = np.nan

# Create meshgrid for axes (match AMPO shape)
```

```
Z, Y, X = np.meshgrid(kjeSet, ftsSet, cfSet, indexing='ij')  # ij indexing: Z=kje, Y=fts, X=cf

# Flatten data for interpolation
x_flat = X.flatten()
y_flat = Y.flatten()
z_flat = Z.flatten()
data_flat = AMPO.flatten()

# Define finer interpolation grid
kje_fine = np.linspace(kjeSet[0], kjeSet[-1], 100)
fts_fine = np.linspace(ftsSet[0], ftsSet[-1], 100)
cf_fine  = np.linspace(cfSet[0], cfSet[-1], 100)

Zf, Yf, Xf = np.meshgrid(kje_fine, fts_fine, cf_fine, indexing='ij')

# Interpolate
AMPO_fine = interpolate.griddata((z_flat, y_flat, x_flat), data_flat, (Zf, Yf, Xf), method='linear

# Find maximum AMPO and corresponding parameters
max_idx = np.nanargmax(AMPO_fine)
iKJE, iFTS, iCF = np.unravel_index(max_idx, AMPO_fine.shape)

opt_kje  = f"{Zf[iKJE, iFTS, iCF]:0.1f}"
opt_fts  = f"{Yf[iKJE, iFTS, iCF]:0.2f}"
opt_cf   = f"{Xf[iKJE, iFTS, iCF]:0.1f}"
maxAMPO = AMPO_fine[iKJE, iFTS, iCF]

# Print results
print(f"Maximum AMPO: {maxAMPO:0.3f}")
print(f"Optimal KJE: {opt_kje} rad, FTS: {opt_fts}, CF: {opt_cf} Hz")
```

Having established that the influence of SSC parameters - cycle frequency, FTS and knee joint excursion - on the maximally attainable AMPO was accurately predicted by our Hill-type MTC model, we now examine the effects of the SSC parameters in detail. The predicted effect of cycle frequency and FTS at four distinct knee joint excursions are depicted in Figure 10. The optimal combination of SSC parameters was predicted to be a cycle frequency of about 1.6 Hz, a FTS of about 0.80 and a knee joint excursion of about 2.0 rad. Because knee joint movements were centred around 1.0 rad, knee joint excursions beyond 2.0 rad would result in knee hyperextension and were therefore not physiologically meaningful. Nevertheless, model predictions showed that knee joint excursions above 2.0 rad did not further increase

the maximally attainable AMPO.

Regarding the interrelationship between the SSC parameters, we found that the optimal FTS for AMPO remained relatively stable across the range of cycle frequencies and knee joint excursions studied, consistently ranging between 0.78 and 0.85. The largest variation in optimal FTS was observed at the smallest knee joint excursion studied (0.8 rad). By contrast, optimal cycle frequency showed substantial sensitivity to changes in FTS. Additionally, a strong interaction was found between cycle frequency and knee joint excursion: increasing one necessitates a decrease in the other to maximise AMPO.

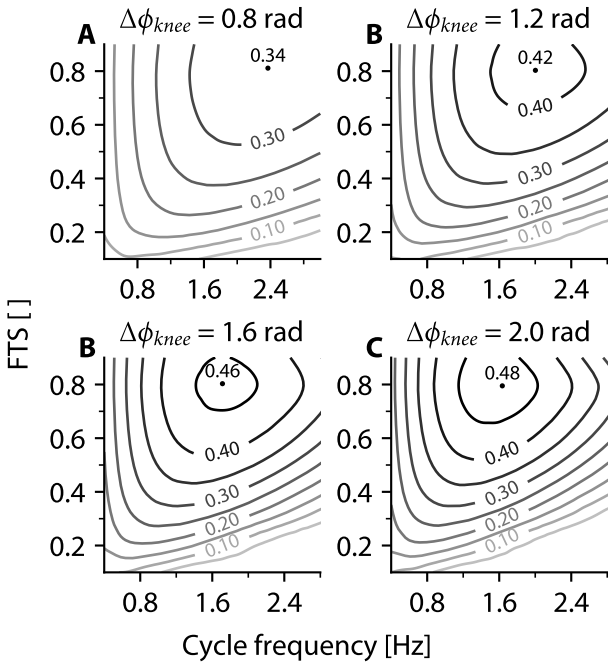

Figure 10: Predicted maximally attainable AMPO as a function of SSC parameters: cycle frequency, fraction of cycle time spent shortening (FTS), and knee joint excursion ($\Delta\phi_{\mathbf{knee}}$). For each subplot, one SSC parameter was fixed, while the influence of the other two parameters on maximally attainable AMPO (normalised to L_{CE}^{opt} and F_{CE}^{max}) is depicted as contour lines. Dots indicate peak AMPO, corresponding to the optimal combination of the two varying SSC parameters; values are shown at the top left of each dot. A–D) Predicted maximally attainable AMPO as a function of cycle frequency and FTS for four distinct knee joint excursions. E–H) Predicted maximally attainable AMPO as a function of cycle frequency and knee joint excursion for four distinct FTS values.

4 Discussion

In this study, we examined the influence of imposed knee joint movement on the maximally attainable AMPO of human *m. quadriceps femoris* by combining experiments with simulations and optimisations using a Hill-type MTC model. Our findings revealed a strong correlation between measured AMPO and predicted maximally attainable AMPO, for a wide variety of imposed knee joint movements. Model predictions revealed that the maximally attainable AMPO peaks at a cycle frequency of about 1.6 Hz, an FTS of about 0.80, and a knee joint excursion of about 2.0 rad. While the optimal knee joint excursion

was predicted to decrease with increasing cycle frequency and vice versa, the optimal FTS was predicted to be largely unaffected by changes in cycle frequency and knee joint excursion.

4.1 Evaluation of model predictions against experimental data

Measured AMPO and predicted maximally attainable AMPO were strongly correlated, with measured AMPO being 87 ± 10% of the predicted values on average. The lower measured AMPO primarily resulted from M_{mus} values that were lower than values predicted for knee extension. It should be noted that, in theory, predicted maximally attainable AMPO should always exceed measured AMPO. For the model predictions, *m. quadriceps femoris* activation was set to be optimal in both timing and magnitude, and no hamstring co-contraction occurs. By contrast, participant muscle activation is inherently suboptimal.

Regarding *m. quadriceps femoris* activation, EMG of *m. quadriceps femoris* rose just at the transition from flexion to extension, causing M_{mus} to increase at this time instant. Thus, we consider the timing of *m. quadriceps femoris* of participants to be adequate. However, EMG of *m. quadriceps femoris* did not exceed 100% of EMG^{MVC} for all conditions amongst five of the six participants. This differs from previous studies (e.g. Bobbert and Harlaar, 1993; Kellis and Baltzopoulos, 1998; Westing et al., 1991), which reported *m. quadriceps femoris* EMG values slightly exceeding 100% in knee dynamometer experiments. Hence, submaximal *m. quadriceps femoris* activation may partly explain the lower M_{mus} relative to predicted values.

Regarding the co-contraction of the hamstring muscle group, EMG of the hamstring muscle group during knee extension was 17.3 ± 12.4% of EMG^{MVC}, consistent with previous studies (e.g. Kellis and Baltzopoulos, 1998; Snow et al., 1993). When interpreting surface EMG as a proxy for muscle activation, potential crosstalk should be considered. In the context of this study, it is possible that both quadriceps and hamstrings activation are slightly overestimated due to crosstalk. However, the effect of EMG crosstalk on the net knee joint moment is probably smaller than suggested by EMG values alone, since the maximal isometric force of the hamstring muscle group is approximately 2.5 times lower than that of *m. quadriceps femoris* (e.g. Prietto and Caiozzo, 1989). Nevertheless, it is clear that co-contraction of hamstrings during knee extension may also partly explain the slight deficit in M_{mus} relative to predicted values.

Measured and predicted M_{mus} were normalised to the maximal isometric force (F_{CE}^{max}; see Figure 8). F_{CE}^{max} was estimated individually for each participant from their measured isometric knee joint angle-moment relationship. In most cases, the actual participant's maximum measured isometric knee joint moment was higher than the maximal value of the model (see Figure 4), indicating that F_{CE}^{max} was not overestimated. Nevertheless, it should be noted that any inaccuracy in F_{CE}^{max} directly affects normalised M_{mus}. Aside from a simple normalisation of AMPO with F_{CE}^{max}, predictions were not individually tailored. Consequently, properties of participants' *m. quadriceps femoris* could be different from the properties used in the Hill-type MTC model. Measured force rise was generally slower than predicted (see Figure 8), suggesting that the model's activation dynamics may have been slightly too fast. Moreover, measured M_{mus} was consistently lower across various knee joint angles, even where maximal activation was

achieved in both experiment and model. The force–length relationship likely played only a minor role, because the isometric knee joint moment–angle relationship was reasonably similar for participants and the model. Instead, the force–velocity relationship appears to be the main contributor. In our model, the decline in knee extension moment with increasing angular velocity was at the lower end of experimentally observed values (for review, see Chen and Franklin, 2025). Overall, the slight deficit in M_{mus} relative to predicted values may partly be explained by differences between the generic model properties and participants' *m. quadriceps femoris* properties. Obviously, all other muscle properties implemented in the model may not exactly match those of (or between) participants' *m. quadriceps femoris*. For example, the moment arm (assumed to be constant in our model) directly influences M_{mus}, yet it was also used to estimate F_{CE}^{max}; thus, any over- or underestimation would be compensated during normalisation. Additionally, the moment arm determines MTC velocity during knee extension, thereby affecting M_{mus} mainly through its influence on CE shortening velocity. Lastly, SEE compliance influences CE dynamics; the additional dynamics resulting from this CE-SEE interaction is captured in our model. Under the present experimental conditions, discrepancies between CE and SEE length and velocity are predicted to be small, as MTC excursions (~6.7 cm) were substantially larger than SEE excursions (~0.6 cm). This suggests that CE–SEE interaction, and therefore any mismatch between modelled and participant-specific SEE compliance, had only a minor influence on M_{mus}. We expect that, with individually tailored models, differences between experimental and predicted M_{mus} would have been even smaller than differences reported in this study.

We emphasise that our aim was not to precisely predict the magnitude of M_{mus} over time, but rather to examine the influence of SSC parameters on the maximally attainable AMPO. Thus, it is promising that we predict the magnitude of the maximally attainable AMPO remarkably well using a relatively simple Hill-type MTC model with generic parameter values obtained from literature, despite the aforementioned methodological limitations. Moreover, when assessing the influence of SSC parameters on the maximally attainable AMPO, the relative change in AMPO is more informative than its (average) magnitude. In this context, it is particularly encouraging that AMPO normalised to z-scores revealed a very strong correlation between predictions and measurements ($r^2 = 0.95$). This finding aligns with previous studies demonstrating that Hill-type MTC models reliably predict mechanical behaviour, especially during concentric contractions (e.g. Ingen Schenau et al., 1988; Lemaire et al., 2016; Sandercock and Heckman, 1997; Wakeling and Johnston, 1999; Zandwijk et al., 1996). Concentric contractions primarily determine maximally attainable AMPO. All in all, we attribute the small discrepancy between measured and predicted maximally attainable AMPO to (a combination of) several factors, such as suboptimal *m. quadriceps femoris* activation, hamstring co-contraction, and generic model assumptions. Nevertheless, predictions aligned remarkably well with measurements, demonstrating the validity of the predicted influence of SSC parameters on the maximally attainable AMPO of human *m. quadriceps femoris*.

4.2 Results agree well with available literature

In the first part of the current study, L_{SEE}^{0}, L_{CE}^{opt} and F_{CE}^{max} were estimated for each participant individually based on isometric measurements at nine different knee joint angles. The measured net knee joint

moment-angle relationship of participants was in line with literature (e.g. Anderson et al., 2007; Eijden et al., 2008; Murray et al., 1980). The estimation of the muscle model parameters was deemed successful as the root mean square error between the measured isometric knee joint moment and the musculoskeletal predicted isometric knee joint moment was low (6.6 ± 2.2% of F_{CE}^{max} on average). Additionally, values of L_{SEE}^{o} and L_{CE}^{opt} corresponded well with values obtained from cadavers (e.g. Friederich and Brand, 1990; Ward et al., 2009; Wickiewicz et al., 1983) and values used in other modelling studies (e.g. Anderson and Pandy, 1999; Soest and Bobbert, 1993). Overall, we conclude that the values of L_{SEE}^{o}, L_{CE}^{opt} and F_{CE}^{max} were successfully estimated.

In the second part of the current study, we examined the maximally attainable AMPO for a large set of imposed knee joint movements. As our study is the first to investigate this for human muscle, the only available data for comparison are those of *in vitro* or *in situ* studies on animal muscle. Whereas we influenced muscle length indirectly by imposing knee joint excursions, those studies directly imposed muscle length excursions. Those studies on animal muscle show that an increase in cycle frequency leads to a decrease in the AMPO-optimal length excursion, muscle stimulation duration and mechanical work per cycle (e.g. Askew and Marsh, 1997; James et al., 1995; Josephson, 1985; Reuvers et al., 2025). Our findings for human *m. quadriceps femoris* align with these observations. Additionally, in accordance with Askew and Marsh (1997) and Reuvers et al. (2025), we found that the optimal (length) excursion increases with increasing FTS and decreasing cycle frequency.

Regarding the values of cycle frequency, FTS and muscle fibre length excursion per se, it should be noted that these values substantially differ due to differences in muscle properties (e.g. maximal shortening velocity, excitation dynamics etc.). Accordingly, the optimal cycle frequency of about 1.6 Hz observed for human *m. quadriceps femoris* contrasts with about 4 Hz for mouse *m. soleus* (Askew and Marsh, 1997), about 9 Hz for mouse *m. extensor digitorum longus* (Askew and Marsh, 1997), and about 3.5 Hz for rat *m. gastrocnemius medialis* (Reuvers et al., 2025). Similarly, optimal muscle fibre length excursion, expressed as a percentage of the optimum fibre length, varies across muscles: 22% for mouse *m. soleus* (Askew and Marsh, 1997), 16% for mouse *m. extensor digitorum longus* (Askew and Marsh, 1997), and 58% for rat *m. gastrocnemius medialis* (Reuvers et al., 2025). All of these values are considerably lower than the 85% reported here for human *m. quadriceps femoris*. Interestingly, the optimal FTS of 0.80 for human *m. quadriceps femoris* is comparable to the range of 0.80-0.90 observed in animal muscle (Askew and Marsh, 1997; Reuvers et al., 2025). The optimal FTS is predominantly a trade-off between (the negative effect of) the force-velocity relationship on the one hand and (the negative effect of) the excitation dynamics on the other hand (for details, see Reuvers et al., 2025). Both factors depend on muscle fibre type; their negative effect on AMPO increases with slower muscle fibre types. Consequently, the optimal FTS is expected to be relatively robust for variations in muscle fibre type. Therefore, it is promising that the optimal FTS in our study aligns with those observed in animal muscle. Overall, our findings are consistent with the available literature, which strengthens our confidence in the validity of the results.

4.3 Experimental constraints and physiological relevance

In extensive pilot experiments, we explored a range of knee joint movements with different angular velocities and accelerations. Based on these pilot experiments and safety considerations, we limited knee joint acceleration to 50 rad/s², thereby also limiting the (maximum achievable) knee joint angular velocity. As a result, only a subset of combinations of SSC parameters — cycle frequency, FTS, and knee joint excursion — could be tested experimentally. Importantly, this does not imply that higher velocities or accelerations are physiologically infeasible for human knee joint movements. In fact, substantially higher values are reached during various motor activities (for review, see Grimmer et al., 2020). For example, during sprint cycling, knee joint angular velocities reach 5–10 rad/s, and angular accelerations exceed 100 rad/s². During sprint running these values are even higher, with knee joint angular velocities reaching about 20 rad/s and angular accelerations up to 200 rad/s². Therefore, the joint movements that were not feasible in our experiment are nonetheless physiologically meaningful.

4.4 Physiological implications for human movement

AMPO in our study was normalised to the product of F_{CE}^{max} and L_{CE}^{opt}. This allowed comparison across participants, but also allows us to extend our findings to muscles of similar fibre types. For typical values of L_{CE}^{opt}=9.3 cm and F_{CE}^{max}=5250 N for *mm. vastii*, and L_{CE}^{opt}=8.1 cm and F_{CE}^{max}=1750 N for *m. rectus femoris* (Soest and Casius, 2000), peak AMPO for *m. quadriceps femoris* is predicted to be approximately 300 W per leg at the optimal combination of SSC parameters. This value may be used as a benchmark in assessing maximum AMPO of human *m. quadriceps femoris* AMPO in any short-duration task that involves periodic motion.

In our study, we measured AMPO of the human *m. quadriceps femoris* during fully imposed knee joint movements. This setup allowed us to directly control the kinematics of the stretch-shortening cycle (SSC) and to ensure that knee joint movements were identical across participants. We show that SSC parameters substantially influence maximally attainable AMPO (see Figure 10). For example, increasing FTS from 0.5 to its optimal value of about 0.8 is predicted to improve AMPO by about 25% at the optimal cycle frequency, with smaller effects at lower frequencies and larger effects at higher frequencies. The resulting insights into the influence of SSC parameters on maximally attainable AMPO can be used to analyse any SSC of any muscle or muscle group, irrespective of whether this SSC arises from fully imposed movements or from unconstrained/self-generated movements. For any SSC, we can now readily estimate the maximal AMPO at the observed kinematics for any muscle (group), and relate that to the maximally attainable AMPO of that muscle (group).

In most real-world situations, movements are not fully constrained and typically involve several mechanical and kinematic degrees of freedom. In such cases, movements arise from muscle forces and mechanical interactions with the environment, which in turn determine the SSCs of the muscles involved. This means that, for example, during an all-out sprint cycling task, the coordination problem is to find muscle activations such that the total AMPO of all muscles combined is maximised over a full cycle. This inevitably leads to a trade-off between the contributions of individual muscles to the

total AMPO. Our results provide not only insight into the mechanics of such a trade-off, but may also help in designing equipment that interacts with the musculoskeletal system to improve AMPO. For example, our results indicate that muscles with a larger volume should spend more time shortening than lengthening compared to muscles with a smaller volume in order to maximise the total AMPO. In many periodic sports, however, the shortening duration of *m. quadriceps femoris* does not exceed the lengthening duration. For instance, FTS ranges from 0.3–0.5 in rowing (Bechard et al., 2009; Ettema et al., 2022), is approximately 0.3 in cross-country skiing (Nilsson et al., 2004; Pellegrini et al., 2014), and is around 0.5 in cycling (Bobbert et al., 2020). Interestingly, in a clever experiment, Martin et al. (2002) showed that during single-leg cycling, the maximal AMPO can be improved by about 4% when the downstroke duration is approximately 10% longer than the upstroke duration. This longer downstroke duration allows the large leg muscles to spend more time shortening than lengthening, leading to an increase in their AMPO that more than offsets the decrease in AMPO of other (smaller) muscles that unavoidably spend less time shortening than lengthening. This example highlight that redesign of equipment that allows muscles to operate closer to their optimal SSC parameters, has the potential to result in improved performance.

4.5 Conclusion

In this study, we examined how SSC parameters — cycle frequency, FTS, and knee joint excursion — influence the maximally attainable AMPO of human *m. quadriceps femoris* during short-duration, all-out (sprint) conditions. Because only a subset of SSC parameters can be tested experimentally, we combined experiments with musculoskeletal modelling. Differences in maximally attainable AMPO were adequately predicted by our Hill-type MTC model, justifying its use to examine the effect of SSC parameters on maximally attainable AMPO in detail. Model predictions showed a substantial influence of SSC parameters on the maximally attainable AMPO. Specifically, predictions revealed a strong interaction between cycle frequency and knee joint excursion: increasing one necessitates a decrease in the other to maximise AMPO. Even more interestingly, in order to maximise AMPO, *m. quadriceps femoris* should spend about 80% of the cycle duration while shortening, independent of cycle frequency and/or knee joint excursion.

Appendix

A1 Selection of experimental conditions

With our musculoskeletal model we predicted the maximally attainable AMPO for knee joint movements with different constant knee angular velocities during flexion and extension. Thus, the knee joint angular velocity changed instantaneously at the switch from flexion to extension (and vice versa), and the knee joint angular acceleration at that instant was infinitely high/low. Clearly, infinite knee joint accelerations cannot be achieved in an experimental setup. Instead, we chose to impose a constant knee angular acceleration until a predefined knee joint angular velocity was achieved — determined by the combination of cycle frequency, FTS, knee joint angle excursion and constant knee angular acceleration — in order to achieve the highest cycle frequency and the greatest difference between maximal and minimal FTS. In extensive pilot tests, we determined that 50 rad/s² was the highest knee angular acceleration that participants were comfortable with and was used as an upper limit for the experiment. Accordingly, the experimental conditions were selected from the knee joint movement that were experimentally feasible (see Figure 3).

Due to the limit on maximal knee joint acceleration, the imposed knee joint movement in the experiment were slightly different from those with a constant knee angular velocity (i.e. with no limit on knee angular acceleration; see Figure A1). As a result, the maximally attainable AMPO differed slightly between those two for the same set of SSC parameters. Therefore, we derived predictions of the maximally attainable AMPO for knee joint movement with a limit of 50 rad/s2 on the knee angular acceleration (see Figure 3).

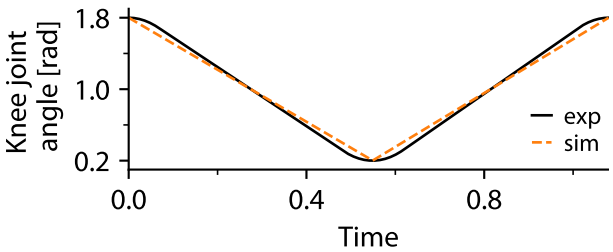

Figure A1: Comparison between knee joint movements during the experiment and simulations. In the experiment, knee joint movements were limited to an angular acceleration of 50 rad/s². In contrast, in the simulations, knee joint movements were used with different constant angular velocities for flexion and extension in order to simulate a broad range knee joint movements.

Among the experimentally feasible knee joint movements, we observed that the influence of cycle frequency and FTS was greatest at highest knee joint excursion. Therefore, we chose to use a knee joint excursion of 1.6 rad, which was the maximum range of motion of the knee dynamometer. Based on the resulting mapping from cycle frequency and FTS at a knee joint excursion of 1.6 rad (see contour lines in Figure 3), we selected two sets of five experimental conditions. One condition was shared between both sets, resulting in nine unique experimental conditions in total. The first set of five conditions was chosen such that the maximally attainable AMPO was predicted to be identical, despite substantial

variations in cycle frequency and FTS (0.35A-E; Figure 3). The second set of conditions was chosen such that the maximally attainable AMPO was predicted to differ substantially due to the imposed variations in cycle frequency and FTS (0.20, 0.25, 0.30, 0.35A-E and 0.40; Figure 3).

A2 Musculoskeletal model

In both our predictions, as well as in the experiments, the knee joint angle over time was fully imposed. Based on the imposed knee joint angle, we calculated the muscle–tendon complex (MTC) length over time using a quadratic relation, as first suggested by Grieve et al. (1978):

$$L_{MTC} = A_0 + A_1 \cdot \phi_{joint} \tag{2}$$

A_0 denotes the muscle-tendon-complex length at a knee joint angle of 0 rad (full extension), with knee joint angle increasing with flexion. ϕ_{joint} denotes the knee joint angle. A_1 defines the change in MTC length based on knee joint angle. MTC mechanical work can be calculated both by $\int F_{MTC} dL_{MTC}$ and $\int M_{MTC} d\phi_{joint} = \int F_{MTC} \cdot r_{MTC} d\phi_{joint}$. An expression for the moment arm of the muscle-tendon-complex around the knee joint (r_{MTC}) can then be derived by substituting these equations into Equation 2:

$$r_{MTC} = \frac{dL_{MTC}}{d\phi_{joint}} = A_1 \tag{3}$$

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B Competing interests

The authors declare no competing or financial interests.

C Author contributions

Conceptualization: E.D.H.M.R., D.A.K., A.J.K.; Formal analysis: E.D.H.M.R.; Investigation: E.D.H.M.R.; Writing - Original Draft: E.D.H.M.R.; Writing - Review & Editing: E.D.H.M.R., D.A.K., A.J.K.; Visualization: E.D.H.M.R.; Supervision: D.A.K., A.J.K.; Funding acquisition: D.A.K., A.J.K.;

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E Data and resource availability

All data, code, and materials used in this study are openly available:

- **GitHub repository:** All raw data, processed data, and analysis code are hosted on GitHub at https://github.com/edwinreuvers/hq-ampo.
- **Reproducible analysis website:** Full analysis pipeline — including data, analysis code, and figure/table generation — is available at https://edwinreuvers.github.io/publications/hq-ampo.

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Description	Symbol	Unit	Value	Reference
				Musculoskeletal geome
MTC length at $\phi_{knee} = 0$	A_0	cm	21.3	van Soest & Casius (2000
Linear coefficient of MTC-ϕ_{knee} relation	A_1	cm/rad	4.20	van Soest & Casius (2000
Hill constant	a^{rel}	-	0.41	van Soest & Casius (2000
				Contraction dynam
Hill constant	b^{rel}	1/s	5.20	van Soest & Casius (2000
Maximum isometric CE force	F_{CE}^{max}	N	5,250	van Soest & Casius (2000
PEE stiffness scaling parameter	k_{PEE}	N/m²	1.19×10^7	van Soest & Bobbert (199
SEE stiffness scaling parameter	k_{SEE}	N/m²	1.28×10^8	van Soest & Casius (2000
CE optimum length	L_{CE}^{opt}	cm	9.30	van Soest & Casius (2000
PEE slack length	L_{PEE}^{o}	cm	13.0	van Soest & Bobbert (199
SEE slack length	L_{SEE}^{o}	cm	16.0	van Soest & Casius (2000
				Excitation dynam
Determines CE force-length relation width	w	-	0.56	van Soest & Casius (2000
Determines steepness γ-q relation	a_{act}	-	-4.59	Hatze (1981)
Determines pCa^{2+} level at which $q = 0.5$	$b_{act,1}$	-	5.17	Hatze (1981)
Determines pCa^{2+} level at which $q = 0.5$	$b_{act,2}$	-	1.08	Hatze (1981)
Determines pCa^{2+} level at which $q = 0.5$	$b_{act,3}$	-	-0.19	Hatze (1981)
Relates γ to pCa^{2+}	kCa	mol/L	8.00×10^{-6}	Kistemaker et al. (2005)
Activation time constant	τ_{act}	ms	88.9	Hatze (1981)
De-activation time constant	τ_{deact}	ms	88.9	Hatze (1981)

Parameter values are depicted only for those key to predict the maximally attainable AMPO.

Table A1: Key muscle parameter values.